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National Yang Ming Chiao Tung University
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二維材料於純量與向量渦旋光激發下之激子躍遷理論研究

Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

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摘　要

中文摘要

關鍵字：純量與向量渦漩光、二維過度金屬二流族化合物、光的自旋-軌道交互作用、谷混合亮激子、灰激子、躍遷速率、波包

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Abstract

The spin angular momentum (SAM) of light was established in 1936, while the orbital angular momentum (OAM) carried by structured light was later demonstrated in 1992 using Laguerre–Gaussian (LG) beams. This additional degree of freedom has significantly enriched research across a wide range of fields. In particular, two–dimensional transition–metal dichalcogenides (TMDs) have emerged as promising platforms for studying light–matter interaction, owing to their finite direct bandgap in the visible to near-infrared spectral range and their strong excitonic effects.

The coupling between SAM and OAM are already found for non-paraxial (focused) light beams. However, only within paraxial approximation can a light beam carried well–defined SAM and OAM. Thus in this study, we focus on finding the the Laguerre–Gaussian (LG) beam

r non-paraxial light beams

Keywords: Scalar and vector vortex beams, two dimensional transition metal dichalcogenides, spin-orbit Interaction of light, valley-mixed bright excitons, gray exciton, transition rate, wave packet

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Chapter 1 Introduction

1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer

Conventional semiconductors such as diamond, silicon [1], and aluminum arsenide are not suitable for the selective initialization, manipulation, and readout of valley states. This is because they possess multiple inequivalent valleys in their band structures that do not couple distinctly to external optical fields. The situation changed with the discovery of graphene in 2004 [2], which triggered rapid progress in research on two-dimensional (2D) materials. Graphene has a hexagonal honeycomb lattice that places two inequivalent valleys at the K and K' points of the hexagonal Brillouin zone. Nevertheless, the absence of a bandgap in graphene allows electrons to be thermally excited between valleys, making valley polarization unstable at room temperature.

In contrast, transition-metal dichalcogenide (TMD) monolayers share a similar honeycomb lattice but possess a finite direct bandgap (see Figure 1.1) on the order of $\sim$ 1–2 eV, which lies in the visible to near-infrared spectral range. A TMD monolayer consists of two group-VIA chalcogen atoms, such as sulfur (S) or selenium (Se), with a group-VIB transition-metal atom, such as molybdenum (Mo) or tungsten (W), located between them.

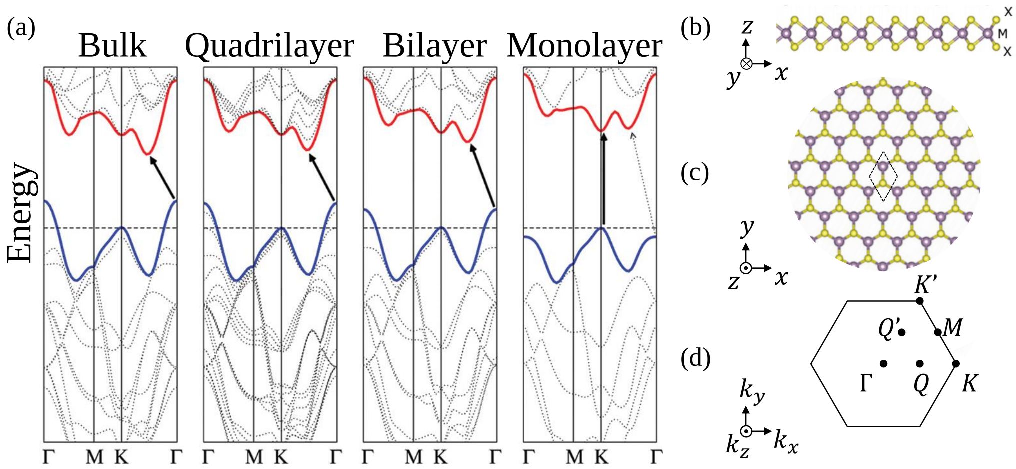

Figure 1.1：(a) Calculated band structures of MoS_2 with different layers, reproduced from Ref. [3]. (b) Side view and (c) top view of transition-metal dichalcogenide (TMD) monolayer with the unit cell framed by the dashed line, reproduced from Ref. [4]. (d) The first Brillouin zone of TMD monolayer in the recipical space and the selected high symetry points

Under optical excitation, electrons in a TMD monolayer can be promoted from the valence band to the conduction band near the K and K' valleys, leaving holes in the valence band. Due to the strong Coulomb interaction in atomically thin materials with weak dielectric screening, the electron−hole pair can bind together to form a quasiparticle known as an exciton. Figure 1.2 shows that center-of-mass (COM) momentum of exciton is related to momentum of light $\boldsymbol{Q}_c$ with incident angle θ through $\boldsymbol{Q} = \boldsymbol{Q}_c \sin\theta$, which gives the idea of the light cone.

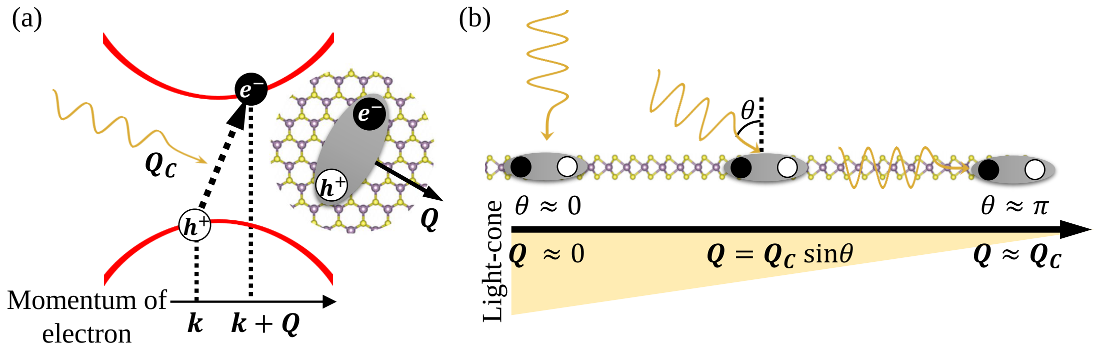

Figure 1.2 : (a) Illustration of a photon carrying momentum $\boldsymbol{Q}_c$ exciting (leaving) an electron (hole) with crystal momentum $\boldsymbol{k} + \boldsymbol{Q}$ $(-\boldsymbol{k})$, forming an exciton with center-of-mass (COM) momentum $\boldsymbol{Q}$. (b) Schematic of ideal of light cone and relation between COM momentum $\boldsymbol{Q}$ and $\boldsymbol{Q}_c$

Moreover, the lack of inversion symmetry in the lattice, together with strong spin−orbit coupling from the heavy transition-metal atoms, gives rise to pronounced spin splitting of the electronic bands. At the K/K' valleys, this splitting is typically on the order of a few meV in the conduction band and several hundreds of meV in the valence band, as illustrated in Figure 1.3(a).

The same symmetry considerations also lead to valley-dependent optical selection rules. As shown in Figure 1.3(b), excitons can be classified according to both momentum and spin selection rules. Only excitons with center-of-mass (COM) momentum $\boldsymbol{Q}$ within the light cone,

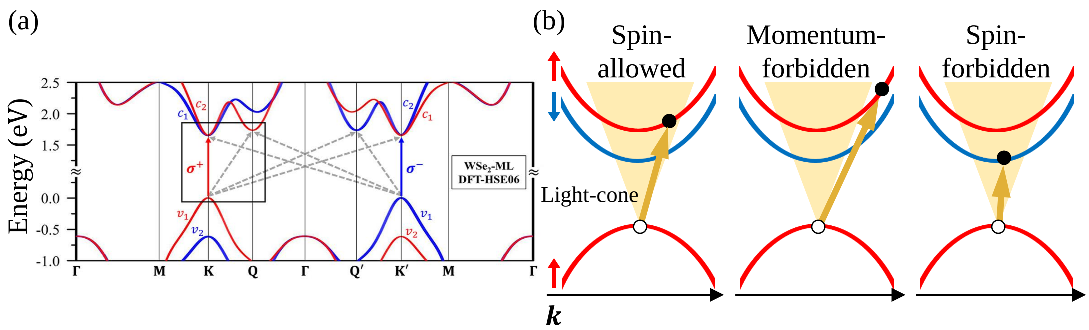

Figure 1.3 : (a) Calculated band structure of WSe_2, showing spin-up (red) and spin-down (blue) states, together with a schematic illustration of valley-selective optical transitions at the K and K' valleys, which couple to right- and left-circularly polarized light, respectively, reproduced from Ref. [5]. (b) Schematic configurations of excitonic states, including spin-allowed bright excitons (BXs), momentum-forbidden dark excitons (DXs), and spin-forbidden DXs.

i.e., $|\boldsymbol{Q}| \lesssim Q_c$, and with antiparallel spin configurations can recombine radiatively and are classified as bright excitons (BXs). In contrast, excitons that lie outside the light cone or violate the spin selection rules remain optically inactive and are therefore classified as dark excitons (DXs).

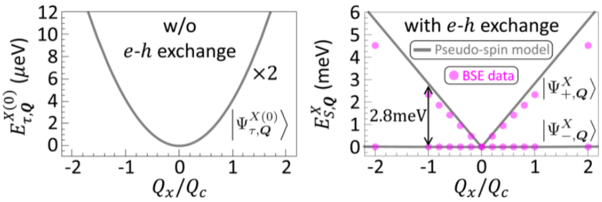

Figure 1.4：The lowest valley-degenerate (left) and valley-split (right) exciton bands in MoS_2 along the Q_x direction, reproduced from Ref. [6].

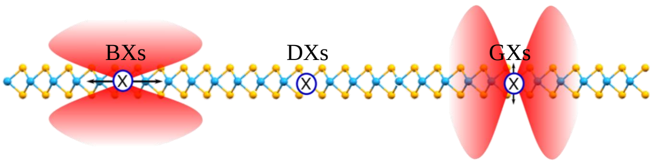

Figure 1.5：Illustration of the radiation patterns of BXs, DXs, and GXs in neutral WSe_2, along with the corresponding optical excitation indicated by arrows (reproduced from Ref. [7]).

The finer band structure further enriches the classification of quasiparticles. In particular, the lowest bright exciton band, which is initially twofold degenerate, splits into two linearly dispersing branches when electron–hole exchange interactions are included within a pseudospin model. These branches are commonly referred to as valley-mixed excitons [6].

In W-based TMDs, an additional excitonic state known as the gray exciton (GX) can also emerge. As shown in Figure 1.5, GXs emit light polarized perpendicular to the two-dimensional plane (out-of-plane direction). Compared with BXs and DXs, GXs exhibit relatively long lifetimes—typically an order of magnitude longer than BXs (BX $\sim$ 10 ps) [7]—while retaining weak but finite optical activity. These properties make GXs particularly useful for studying exciton dynamics and light–matter interactions, and highlight the rich excitonic physics in two-dimensional TMDs.

1.2 Spin and Orbital Angular Momentum (SAM and OAM) of Laguerre–Gaussian (LG) beam

Following the theoretical work of Poynting in 1909 [8] and the experimental demonstration by Beth in 1936 [9], it has been established that each photon in circularly polarized light with finite dimensions [10–12] carries a spin angular momentum (SAM) of $S = \sigma\hbar = \pm\hbar$ along the propagation direction, where σ denotes the photon helicity. In Beth's experiment, the torque exerted on a suspended half-wave plate by circularly polarized light was measured, demonstrating that spin angular momentum can be transferred to a birefringent medium during propagation (see Figure 1.6(a)).

In 1992, Allen *et al.* showed that a laser beam in a Laguerre–Gaussian (LG) mode possesses a well-defined orbital angular momentum (OAM), also directed along the propagation axis, with $L = \ell\hbar$ per photon, where $\ell \in \mathbb{Z}$ is the azimuthal mode index, commonly referred to as the topological charge [10]. In their experiment, an input Hermite–Gaussian mode, which carries zero OAM, was converted into an LG mode with nonzero OAM. This mode conversion subsequently induced a measurable torque on cylindrical optical elements (see Figure 1.6(b, c)), establishing that light can also carry OAM associated with the spatial structure of its wavefront. The helical phase structure associated with a quantized amount of $\ell\hbar$ of OAM is illustrated in Figure 1.7, while intensified charge-coupled device (ICCD) images show the characteristic doughnut-shaped intensity distribution.

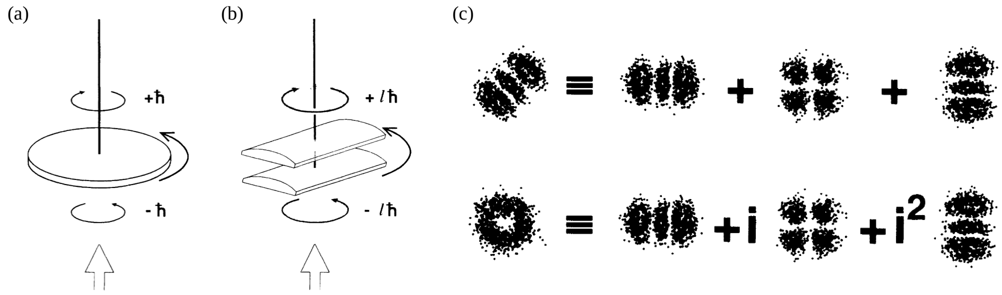

Figure 1.6：(a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre–Gaussian (LG) mode carrying OAM $\ell\hbar$ per photon into a mode with $-\ell\hbar$ per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero ℓ (left side of bottom row) by introducing a phase shift when converting Hermite–Gaussian modes with zero ℓ (left side of top row). All reproduced from Ref. [10].

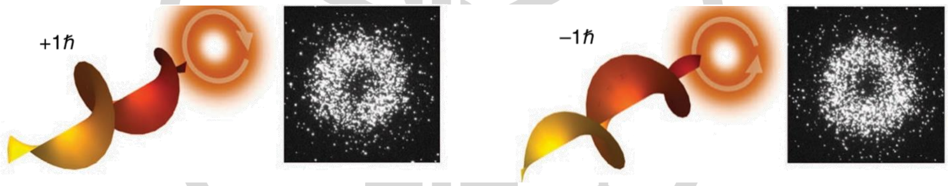

Figure 1.7：Illustration of twisted light with a helical wavefront, where each photon carries $\ell\hbar$ of orbital angular momentum. The intensified charge-coupled device camera images show the characteristic doughnut-shaped intensity profile. Reproduced from Ref. [13].

Since then, a wide range of applications of light carrying OAM has been developed. Early studies focused on angular momentum transfer from light to matter, where objects experience mechanical torque due to OAM transfer, resulting in rotational motion [14–16]. Later developments have extended to optical communications [17–19], quantum key distribution [20, 21], astronomy [22], data storage [23, 24], enhanced imaging sensitivity [25, 26], and optical tweezers [14, 27]. These advances motivate investigation of how structured light carrying orbital angular momentum interacts with electronic excitations in low-dimensional materials.

1.3 SAM and OAM coupling in LG beam

In free space, coupling between SAM and OAM has been shown to occur in non-paraxial (tightly focused) light. Along the propagation direction, SAM can be partially converted into

OAM as the focusing angle increases [28]. This effect can be understood by considering the reduction of the spin component along the beam axis induced by a focusing lens (see Figure 1.8).

The initial spin s_0 is aligned with the propagation direction. After focusing, the spin is no longer parallel to the beam axis; instead, the final spin s_1 is tilted, so only its longitudinal component s_z contributes to the beam’s spin angular momentum. Since the maximum spin angular momentum is $s\hbar$ per photon, the total spin is reduced. However, total angular momentum is conserved during focusing. Consequently, this reduction in SAM is compensated by orbital angular momentum generation, leading to a transfer from SAM to OAM [29].

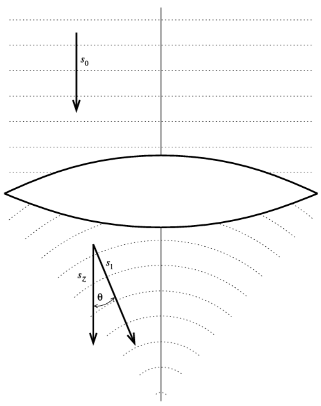

Figure 1.8：Illustration of a beam focused by a lens, showing the variation of SAM before and after the lens along the propagation direction. Reproduced from Ref. [29].

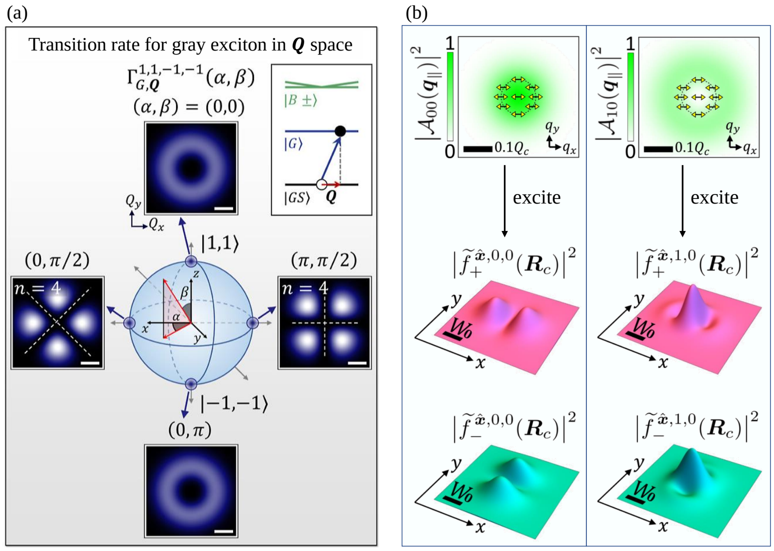

Figure 1.9：(a) Transition-rate distribution for gray excitons in momentum space, reproduced from Ref. [30]. (b) Distribution of the incident light and the excited BX wave-packet states, reproduced from Ref. [6].

However, it should be emphasized that the simple proportionality between spin and orbital angular momentum with the optical helicity σ and topological charge ℓ holds strictly only within the paraxial approximation [31, 32], which often neglects the longitudinal component. However, it is precisely this longitudinal component that can couple to GXs. More importantly, this neglected component already contains two distinct types of spin–orbit interaction (SOI).

In 2024, Oscar Javier Gomez Sanchez *et al.* demonstrated that a type of SOI originating from the phase of the longitudinal component induces momentum-dependent transition rates of GXs through the superposition of two twisted-light beams, forming the vector vortex beams (VVBs) (see Figure 1.9(a)) [30]. However, the combined contribution of SAM and OAM was not fully clarified in their formulation. Meanwhile, in 2022, Guan-Hao Peng *et al.* investigated the photoexcitation of valley-mixed BXs by LG beams within the pseudospin model, suggesting a connection to excitonic topology [6] (see Figure 1.9(b)). Nevertheless, the global phase of the valley-mixed excitons may still influence the excited wave-packet profile.

In this work, we extend the studies of Oscar Javier Gomez Sanchez *et al.* and Guan-Hao Peng *et al.* to investigate the interplay between spin and orbital angular momentum of light in determining transition rates and wave-packet distributions of excitonic states within SI units.

Chapter 2 SAM and OAM interaction (SOI) of LG Beam

2.1 Gauge Restriction

The derivation of Laguerre–Gaussian (LG) beams involves a series of approximations, among which the paraxial approximation plays a central role. Importantly, the wave equation obtained under the paraxial approximation is not independent of the choice of gauge; rather, it is implicitly constrained by the adopted gauge condition.

In this section, we examine how gauge choice restricts the formulation of wave equations. We begin by introducing the wave equations under different gauge conditions, and then analyze the compatibility of the paraxial approximation. This discussion clarifies why certain gauges are more suitable than others for describing paraxial optical fields.

2.1.1 Gauge Freedom in Wave Equations

Maxwell's equations describe how the **electric field** $\boldsymbol{E}(\boldsymbol{r}, t)$ and the **magnetic field** $\boldsymbol{B}(\boldsymbol{r}, t)$ evolve in space and time. These equations form the fundamental basis for describing electromagnetic waves, including Laguerre–Gaussian beams. In **free space**, the charge density ρ and current density $\mathbf{J}$ vanish. In this case, Maxwell's equations reduce to four homogeneous differential equations involving the vacuum permittivity ϵ_0, the vacuum permeability μ_0, and the speed of light in vacuum $c = 1/\sqrt{\mu_0 \epsilon_0}$.

Within this electrodynamics framework, Maxwell's equations can be rewritten as two coupled wave equations for the **scalar and vector potentials**, $\phi(\boldsymbol{r}, t)$ and $\boldsymbol{A}(\boldsymbol{r}, t)$, derived in appendix A.1 as

$$\nabla^2 \phi(\boldsymbol{r}, t) + \frac{\partial}{\partial t} \left(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t) \right) = 0, \tag{2.1}$$

$$\nabla^2 \boldsymbol{A}(\boldsymbol{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \boldsymbol{A}(\boldsymbol{r}, t) - \boldsymbol{\nabla} \left(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\boldsymbol{r}, t) \right) = 0. \tag{2.2}$$

These equations contain the same physical information as the original Maxwell equations, but are expressed in terms of the scalar and vector potentials.

Further simplification of Eqs. (2.1) and (2.2) can be achieved by exploiting the gauge freedom of the potentials. Below, we introduce two commonly used gauges: the **Coulomb gauge** and the **Lorenz gauge**. Superscripts C and L are used to distinguish quantities defined in the

Coulomb and Lorenz gauges, respectively. The gauge conditions are given by

$$\text{Coulomb gauge condition}: \quad \boldsymbol{\nabla} \cdot \boldsymbol{A}^C(\boldsymbol{r}, t) = 0, \tag{2.3}$$

$$\text{Lorenz gauge condition}: \quad \boldsymbol{\nabla} \cdot \boldsymbol{A}^L(\boldsymbol{r}, t) = -\frac{1}{c^2}\frac{\partial}{\partial t}\phi^L(\boldsymbol{r}, t). \tag{2.4}$$

Both gauges simplify the wave equations. In the Coulomb gauge, they reduce to

$$\nabla^2 \phi^C(\boldsymbol{r}, t) = 0, \text{ and} \tag{2.5}$$

$$\nabla^2 \boldsymbol{A}^C(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}^C(\boldsymbol{r}, t) = \frac{1}{c^2}\frac{\partial}{\partial t}\boldsymbol{\nabla}\phi^C(\boldsymbol{r}, t) = 0, \tag{2.6}$$

where the last equality follows from Eq. (2.5). In the Lorenz gauge, Eqs. (2.1) and (2.2) become

$$\nabla^2 \phi^L(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi^L(\boldsymbol{r}, t) = 0, \text{ and} \tag{2.7}$$

$$\nabla^2 \boldsymbol{A}^L(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}^L(\boldsymbol{r}, t) = 0. \tag{2.8}$$

The physical field $\boldsymbol{B}(\boldsymbol{r}, t)$ and $\boldsymbol{E}(\boldsymbol{r}, t)$ can be related to the vector potentials defined in different gauges through Eq. (A.1.5) and

$$\boldsymbol{E}(\boldsymbol{r}) = i\omega \boldsymbol{A}^C(\boldsymbol{r}) = i\omega\left(\boldsymbol{A}^L(\boldsymbol{r}) + \frac{\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}^L(\boldsymbol{r})\right)}{q_0^2}\right). \tag{2.9}$$

The derivation of this relation assumes a **monochromatic wave** ansatz. Specifically, the fields and potentials are written as $\boldsymbol{E}(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{E}(\boldsymbol{r})e^{-i\omega t}\right]$, $\boldsymbol{A}^C(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^C(\boldsymbol{r})e^{-i\omega t}\right]$, $\boldsymbol{A}^L(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\right]$, and $\phi^L(\boldsymbol{r}, t) = \text{Re}\left[\phi^L(\boldsymbol{r})e^{-i\omega t}\right]$, where ω is the angular frequency of light. Substituting these expressions into Eq. (A.1.6) and applying the corresponding gauge conditions leads to Eq. (2.9). Here $q_0 = \omega/c$ denotes the wavenumber of light in free space.

2.1.2 Incompatibility of Gauge Freedom with the Paraxial Approximation

In principle, the vector potential formulation of LG beams can be derived starting from either Eq. (2.6) or Eq. (2.8). However, only the Lorenz gauge remains fully consistent throughout the derivation, whereas the Coulomb gauge condition becomes incompatible with the paraxial wave ansatz. For this reason, we adopt the Lorenz gauge in the following derivation for simplicity and consistency, and verify this choice at the end.

Before applying the paraxial approximation, the scalar Helmholtz equation is obtained from Eq. (2.8) by imposing both the monochromatic ($\boldsymbol{A}^L(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\right]$) and transverse wave ansätze ($\boldsymbol{A}^L(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}} A^L(\boldsymbol{r})$, where $\hat{\boldsymbol{\varepsilon}}$ is a polarization unit vector) [33] within the Lorenz gauge (see Sec. A.2):

$$\nabla^2 A^L(\boldsymbol{r}) + q_0^2 A^L(\boldsymbol{r}) = 0. \tag{2.10}$$

The **paraxial approximation** [34–36] assumes that the rays have only small inclinations with respect to the optical axis in geometric optics. In wave optics, this corresponds to an angular spectrum dominated by plane-wave components whose wavevectors are close to the propagation direction. Here, we choose the propagation direction along the $\boldsymbol{z}$ axis, such that the dominant component of the wavevector $\boldsymbol{q}_0$ lies along $\hat{\boldsymbol{z}}$. Under this approximation, we introduce the **paraxial wave** ansatz, $A^L(\boldsymbol{r}) = U(\boldsymbol{r})e^{iq_0 z}$, and substitute it into Eq. (2.10), which yields

$$\nabla_{\parallel}^2 U(\boldsymbol{r}) + \frac{\partial^2}{\partial z^2} U(\boldsymbol{r}) + 2iq_0 \frac{\partial}{\partial z} U(\boldsymbol{r}) = 0. \tag{2.11}$$

Here, $\nabla_{\parallel}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ denotes the transverse Laplacian. The paraxial wave ansatz assumes that the **envelope function** $U(\boldsymbol{r})$ **varies slowly along the** $\boldsymbol{z}$ **direction** compared to the rapid phase variation of $e^{iq_0 z}$. Consequently, the paraxial approximation implies the following relations:

$$\left| \frac{\partial^2}{\partial z^2} U(\boldsymbol{r}) \right| \ll \left| \nabla_{\parallel}^2 U(\boldsymbol{r}) \right|, \qquad \left| \frac{\partial^2}{\partial z^2} U(\boldsymbol{r}) \right| \ll q_0 \left| \frac{\partial}{\partial z} U(\boldsymbol{r}) \right|.$$

Neglecting the second-order derivative with respect to z then leads to the paraxial Helmholtz equation

$$\nabla_{\parallel}^2 U(\boldsymbol{r}) + 2iq_0 \frac{\partial}{\partial z} U(\boldsymbol{r}) = 0. \tag{2.12}$$

We now clarify why the Coulomb gauge is not suitable in this derivation. The paraxial wave ansatz generally does not satisfy the Coulomb gauge condition (Eq. (2.3)), which requires

$$\boldsymbol{\nabla} \cdot \boldsymbol{A}^C(\boldsymbol{r}, t) = \hat{\boldsymbol{\varepsilon}} \cdot \boldsymbol{\nabla} A^C(\boldsymbol{r}, t) = 0. \tag{2.13}$$

For a transverse wave with the polarization unit vector $\hat{\varepsilon}$ confined to the x–y plane [33], this condition reduces to

$$\hat{\boldsymbol{\varepsilon}} \cdot \boldsymbol{\nabla}_{\parallel} A^C(\boldsymbol{r}, t) = 0, \tag{2.14}$$

where $\boldsymbol{\nabla}_{\parallel} = \frac{\partial}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}\hat{\boldsymbol{y}}$. This condition constrains the transverse spatial variation of $A^C(\boldsymbol{r}, t)$ along the polarization direction, which is incompatible with the assumed slowly varying envelope function $U(\boldsymbol{r})$ [34]. In other words, the Coulomb gauge imposes an additional constraint that conflicts with the paraxial wave ansatz. To satisfy the divergence-free condition, a nonzero longitudinal (z) component of the vector potential must be introduced, going beyond the simple transverse ansatz. Therefore, the Lorenz gauge provides a more natural and consistent framework for deriving the paraxial Helmholtz equation and LG beams solutions.

2.2 Manifestation of SOI in LG Beam

Within the paraxial approximation, the optical helicity σ and the topological charge ℓ are directly proportional to the spin and orbital angular momentum of light, respectively [31, 32]. This correspondence provides a convenient framework for analyzing the angular momentum properties of LG beams.

Moreover, when the vector potential formulation of LG beams are expressed in the Coulomb gauge, an SOI can emerge through the additional longitudinal component of the field, giving rise to terms proportional to $\sigma\ell$ and $\sigma + \ell$. Adopting the Coulomb gauge is further justified by the theory of light−matter interaction, where the vector potential is conventionally formulated within both the Coulomb gauge and the angular spectrum representation. In this representation, approximately interpret the in-plane wavevector as a vector in reciprocal space enable a connection to theory of matter.

In the following section, we first introduce the LG beams solutions in both real-space and angular spectrum representations. We then perform a gauge transformation to reveal the SOI.

2.2.1 Solutions of LG Beams

Under the **long Rayleigh-length range** [37], the LG beams in real space can be solved from Eq. (2.12) within cylindrical coordinates as

$$\boldsymbol{A}_{\boldsymbol{q}_0}^{\hat{\boldsymbol{\varepsilon}},\ell,p;L}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}} A_0^{\mathrm{LG}}\, u_{\ell,p}(\boldsymbol{r}_\parallel)\, e^{iq_0 z}, \tag{2.15}$$

where the position vector is written as $\boldsymbol{r} = (\rho, \phi, z) = (\boldsymbol{r}_\parallel, z)$. Here $\boldsymbol{r}_\parallel$ denotes the in-plane position vector consisting of the radial coordinate ρ and the azimuthal angle ϕ. The integer $\ell \in \mathbb{Z}$ is the **topological charge**, and the integer $p \geq 0$ is the radial mode index. The vector $\boldsymbol{q}_0 = q_0\hat{\boldsymbol{z}}$ denotes the wavevector along the propagation direction, and A_0^{LG} is a constant amplitude. The encelope function is expressed as

$$U(\rho, \phi, z) \approx u_{\ell,p}(\boldsymbol{r}_\parallel) = f_{\ell,p}(\rho)\, e^{i\ell\phi}, \tag{2.16}$$

where the radial function is given by [6, 30, 37] $f_{\ell,p}(\rho) = \sqrt{\frac{2p!}{\pi(|\ell|+p)!}}\, L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right)$. Here the generalized Laguerre polynomial is defined as $L_p^{|\ell|}(x) = \sum_{m=0}^{p}(-1)^m \frac{(|\ell|+p)!}{(p-m)!(|\ell|+m)!m!}x^m$, with $x = \frac{2\rho^2}{W_0^2}$. The parameter W_0 denotes the beam waist of the LG beam, while the cylindrical coordinates are related to Cartesian coordinates through $\rho = \sqrt{x^2 + y^2}$ and $\phi = \tan^{-1}(y/x)$.

Focusing on the minimum cross-section area of the beam ($z = 0$), Figure 2.1 shows that as ℓ and/or p increase, $u_{\ell,p}(\boldsymbol{r}_\parallel)$ develops a broader doughnut-shaped intensity profile, accompanied by an increasing number of radial nodes along the radial direction. The phase varies continuously with the azimuthal angle ϕ, reflecting the helical wavefront associated with orbital angular momentum. In particular, the phase winds by $2\pi\ell$ around the beam center. The beam waist is chosen as $W_0 = 1.5\ \mu$m, which sets the spatial scale of the beam and determines the radial extent of the intensity distribution.

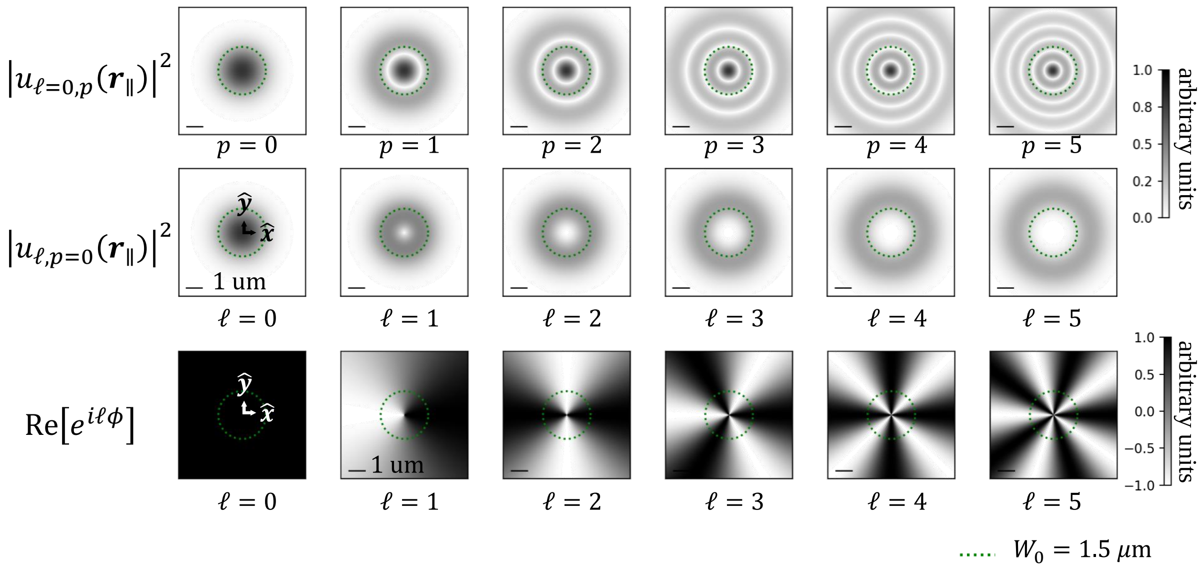

Figure 2.1：Squared magnitude (intensity) and phase distribution of $u_{\ell,p}(\boldsymbol{r}_\parallel)$ at $z = 0$. The green dotted line denotes the beam waist W_0. The scale bar indicates the length of 1 um.

To derive the vector potential of LG beams in the angular spectrum representation, the field is expanded in a plane-wave basis as

$$\boldsymbol{A}(\boldsymbol{r}) = \sum_{\boldsymbol{q}} \boldsymbol{\mathcal{A}}(\boldsymbol{q}) e^{i\boldsymbol{q}\cdot\boldsymbol{r}}, \tag{2.17}$$

where $\boldsymbol{\mathcal{A}}(\boldsymbol{q})$ is the complex amplitude associated with each plane-wave component $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$. The LG beams in Eq. (2.15) can be written in this form through a defined **two-dimensional Fourier transformation** and its corresponding inverse transformation,

$$u_{\ell,p}(\boldsymbol{r}_\parallel) = \frac{1}{(2\pi)^2}\int \mathrm{d}^2\boldsymbol{q}_\parallel\, F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel)\, e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}, \quad \longleftrightarrow \quad F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel) = \int \mathrm{d}^2\boldsymbol{r}_\parallel\, u_{\ell,p}(\boldsymbol{r}_\parallel)\, e^{-i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}. \tag{2.18}$$

For applications to two-dimensional materials, the in-plane wavevector $\boldsymbol{q}_\parallel$ can be approximately regarded as a vector in reciprocal space. Under this approximation, Eq. (2.18) can be discretized, and the Eq. (2.17) becomes

$$\boldsymbol{A}^{\hat{\varepsilon},\ell,p;L}_{\boldsymbol{q}_0}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}} \sum_{\boldsymbol{q}_\parallel} \mathcal{A}^{L}_{\ell,p}(\boldsymbol{q}_\parallel) e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel} e^{iq_0 z}. \tag{2.19}$$

Here the in-plane wavevector is written as $\boldsymbol{q}_\parallel = (q_\parallel, \phi_{\boldsymbol{q}_\parallel})$. Since the spatial distribution has the form $u_{\ell,p}(\boldsymbol{r}_\parallel) = f_{\ell,p}(\rho)e^{i\ell\phi}$, the complex amplitude can be expressed analytically as

$$\mathcal{A}^{L}_{\ell,p}(\boldsymbol{q}_\parallel) = \tilde{F}_{\ell,p}(q_\parallel) e^{i\ell\phi_{\boldsymbol{q}_\parallel}}, \tag{2.20}$$

where $\tilde{F}_{\ell,p}(q_\parallel) = \frac{A_0^{\mathrm{LG}}}{\Omega}(-1)^{|\ell|}2\pi\, \mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)]$, and $\mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)] = \int_0^\infty \mathrm{d}\rho\ \rho\, f_{\ell,p}(\rho)\ J_{|\ell|}(q_\parallel\rho)$ is the **Hankel transform** of order $|\ell|$, with $J_{|\ell|}(q_\parallel\rho)$ being the **Bessel function of the first kind**. Here Ω denotes the area of the two-dimensional system.

In this study, we focus on the fundamental radial mode $p = 0$. In real space, the vector potentials take the explicit form

$$\boldsymbol{A}_{\boldsymbol{q}_0}^{\hat{\varepsilon},\ell,p=0;L}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}} \frac{A_0^{\mathrm{LG}}}{\sqrt{|\ell|!\,\pi/2}} \left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right) \exp\left[i(\ell\phi + q_0 z)\right], \tag{2.21}$$

whereas in the angular spectrum representation the amplitude become

$$\mathcal{A}_{\ell,p=0}^{L}(\boldsymbol{q}_\parallel) = \left(\frac{\pi A_0^{\mathrm{LG}} W_0^2 (-1)^{|\ell|}}{\Omega\sqrt{|\ell|!\,\pi/2}}\right) \left[\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|} \exp\left(-\frac{q_\parallel^2 W_0^2}{4}\right)\right] \exp\left(i\ell\phi_{\boldsymbol{q}_\parallel}\right). \tag{2.22}$$

Similarly, focusing on the $z = 0$ plane, Figure 2.2 shows that $\mathcal{A}_{\ell,p=0}^{L}(\boldsymbol{q}_\parallel)$ exhibits an intensity profile analogous to that of $u_{\ell,p=0}(\boldsymbol{r}_\parallel)$, reflecting the characteristic doughnut-shaped structure in angular spectrum. This correspondence follows directly from the Fourier transform relation between two representations of the LG beam. The doughnut-shaped distribution in angular spectrum indicates that the dominant in-plane wavevectors lie within a finite range determined by the beam waist W_0.

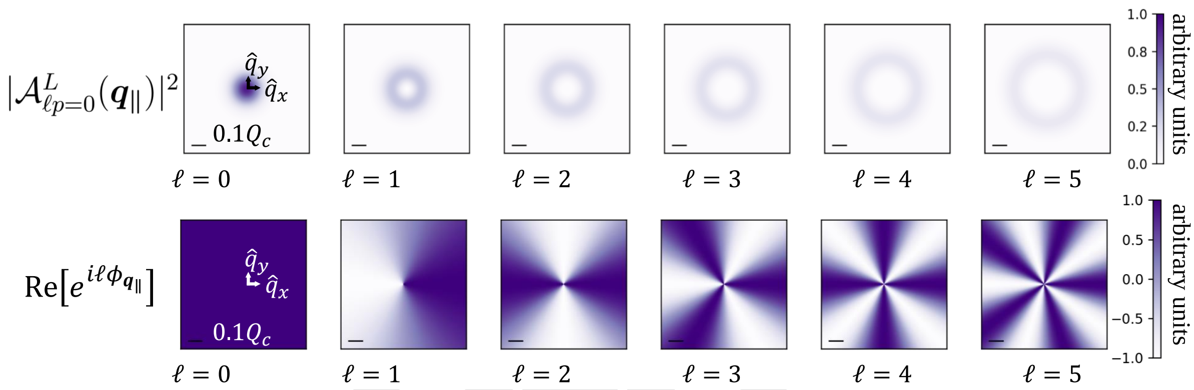

Figure 2.2：Squared magnitude and phase distribution of $\mathcal{A}_{\ell,p=0}^{L}(\boldsymbol{q}_\parallel)$ at $z = 0$. The scale bar indicates the considered 0.1 times light cone range for WSe_2-monolayers, $Q_c = 8.6\ \mu\mathrm{m}^{-1}$.

2.2.2 SOI in the Phase of LG beam's z Component

In the previous section, Eq. (2.9) was introduced to relate the vector potentials defined in the two gauges to the physical fields. Using this relation, we can perform gauge transformation by rewritten Eq. (2.9) as

$$\boldsymbol{A}_{\boldsymbol{q}_0}^{\hat{\varepsilon},\ell,p;C}(\boldsymbol{r}) = \boldsymbol{A}_{\boldsymbol{q}_0}^{\hat{\varepsilon},\ell,p;L}(\boldsymbol{r}) + \frac{\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot \boldsymbol{A}_{\boldsymbol{q}_0}^{\hat{\varepsilon},\ell,p;L}(\boldsymbol{r})\right)}{q_0^2}. \tag{2.23}$$

To explicitly incorporate the SAM of light, the polarization unit vector is defined as $\hat{\boldsymbol{\varepsilon}} = \frac{1}{\sqrt{2}}(\hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}})$, where the photon helicity $\sigma = \pm 1$ denotes right- and left-handed circular polarizations. For LG beams in the fundamental radial mode $p = 0$, the vector potential in real space under the

Coulomb gauge is obtained as

$$\boldsymbol{A}^{\hat{\varepsilon},\ell,p=0;C}_{\boldsymbol{q}_0}(\boldsymbol{r}) \approx A_0^{\mathrm{LG}} \left[f_{\ell,p=0}(\rho)e^{i\ell\phi}\hat{\boldsymbol{\varepsilon}} + \frac{i}{\sqrt{2}q_0\rho}\left((|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right) f_{\ell,p=0}(\rho)e^{i(\sigma+\ell)\phi}\hat{\boldsymbol{z}} \right] e^{iq_0 z}. \tag{2.24}$$

In general, the second term in Eq. (2.23) introduces corrections to all spatial components of the vector potential. However, under the paraxial approximation, these corrections are suppressed. Therefore, we retain only the longitudinal (z) component as the leading correction.

The longitudinal component in Eq. (2.24) reveals two distinct types of SOI: a $\sigma\ell$ dependence in the amplitude and a $\sigma+\ell$ dependence in the phase. Figure 2.3 shows that the magnitude (absolute value) at $z = 0$ of this longitudinal component is significantly weaker than that of the transverse components. The $\sigma\ell$ dependence manifests as an asymmetry between cases with the same ℓ but opposite helicities σ.

It is worth noting that the longitudinal term is weighted by a factor proportional to $(q_0W_0^2)^{-1}$, which is inversely related to the Rayleigh range ($\frac{1}{2}q_0W_0^2$) [35] . This indicates that a smaller beam waist W_0 leads to a stronger longitudinal component, corresponding to a shorter Rayleigh range and tighter focusing, as shown in the bottom row of Figure 2.3. The beam waist W_0 is chosen to be 1.5 μm and 0.5 μm, representing a typical paraxial beam and a case at the boundary of the paraxial regime, respectively [38–40]. This demonstrates that the longitudinal component, although small, remains finite even within the paraxial regime.

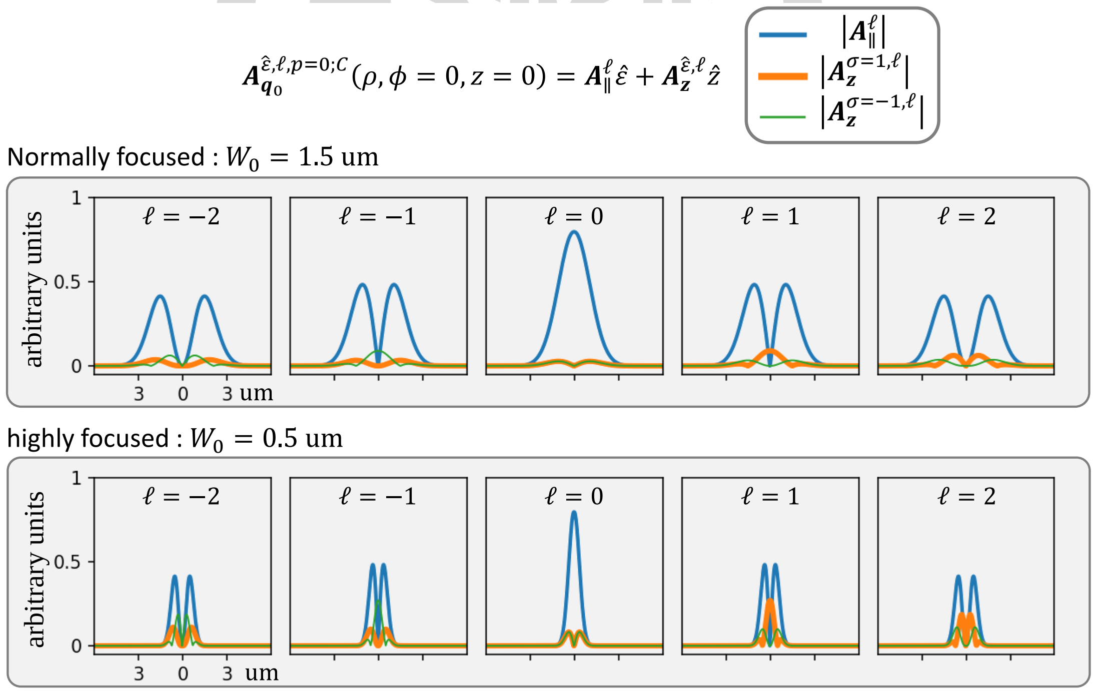

Figure 2.3 : Magnitude (absolute value) of $\boldsymbol{A}^{\hat{\varepsilon},\ell,p=0;C}_{\boldsymbol{q}_0}(\boldsymbol{r})$ at $z = 0$ for different components and helicities. The $\sigma\ell$ term leads to asymmetry between cases with the same ℓ in the longitudinal component (a wavelength of 532 nm is assumed).

As for the $\sigma + \ell$ dependence in the phase term, it is retained in the angular spectrum representation and will be discussed below. By combining Eq. (2.23) with the two-dimensional

Fourier transformation defined in Coulomb gauge, vector potential can be written as

$$\boldsymbol{A}_{\boldsymbol{q}_0}^{\hat{\varepsilon},\ell,p=0;C}(\boldsymbol{r}) = \frac{1}{(2\pi)^2}\int \mathrm{d}^2\boldsymbol{q}_{\parallel}\, \boldsymbol{\mathcal{A}}_{\ell,p}^{C}(\boldsymbol{q}_{\parallel}) e^{i\boldsymbol{q}_{\parallel}\cdot\boldsymbol{r}_{\parallel}} e^{iq_0 z}. \tag{2.25}$$

The corresponding complex amplitude in the angular spectrum representation is obtained as

$$\begin{aligned}
\boldsymbol{\mathcal{A}}_{\ell,p}^{C}(\boldsymbol{q}_{\parallel}) &= \left[\hat{\varepsilon} - \left(\frac{|\boldsymbol{q}|}{q_0}\right)^2 (\hat{\varepsilon}\cdot\hat{\boldsymbol{q}}_{\parallel})\hat{\boldsymbol{q}}\right] \mathcal{A}_{\ell,p}^{L}(\boldsymbol{q}_{\parallel}) && (2.26)\\
&= \left[\hat{\varepsilon} - \left(\frac{|\boldsymbol{q}|}{q_0}\right)^2 \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\sigma\phi_{\boldsymbol{q}_{\parallel}}}\hat{\boldsymbol{q}}\right] \mathcal{A}_{\ell,p}^{L}(\boldsymbol{q}_{\parallel}) && (2.27)\\
&\approx \left[\hat{\varepsilon} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\sigma\phi_{\boldsymbol{q}_{\parallel}}}\hat{\boldsymbol{z}}\right] \mathcal{A}_{\ell,p}^{L}(\boldsymbol{q}_{\parallel}), && (2.28)
\end{aligned}$$

where the last approximation follows from the paraxial condition $\frac{q_0}{|\boldsymbol{q}|} = \cos\theta_{\boldsymbol{q}} \approx 1$. The unit vector is given by $\hat{\boldsymbol{q}} = \sin\theta_{\boldsymbol{q}}\left(\cos\phi_{\boldsymbol{q}_{\parallel}}\hat{\boldsymbol{x}} + \sin\phi_{\boldsymbol{q}_{\parallel}}\hat{\boldsymbol{y}}\right) + \cos\theta_{\boldsymbol{q}}\hat{\boldsymbol{z}} \approx \hat{\boldsymbol{z}}$, and $\sin\theta_{\boldsymbol{q}} = \frac{q_{\parallel}}{\sqrt{q_{\parallel}^2+q_0^2}}$. Substituting the Lorenz gauge amplitude obtained previously, the Coulomb gauge complex amplitude in the angular spectrum representation becomes

$$\boldsymbol{\mathcal{A}}_{\ell,p}^{C}(\boldsymbol{q}_{\parallel}) \approx \tilde{F}_{\ell,p}(q_{\parallel}) e^{i\ell\phi_{\boldsymbol{q}_{\parallel}}}\hat{\boldsymbol{\varepsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{\ell,p}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{\boldsymbol{q}_{\parallel}}}\hat{\boldsymbol{z}}. \tag{2.29}$$

Notably, the SOI remains encoded in the phase of the longitudinal component through the $\sigma+\ell$ dependence, corresponding to the total angular momentum (TAM), as illustrated in Figure 2.4. However, this phase contribution is generally difficult to detect experimentally, as it tends to average out in intensity-based measurements. To access this TAM contribution, one can employ the interference of two LG beams, forming so-called vector vortex beams (VVBs), which will be discussed in the following section.

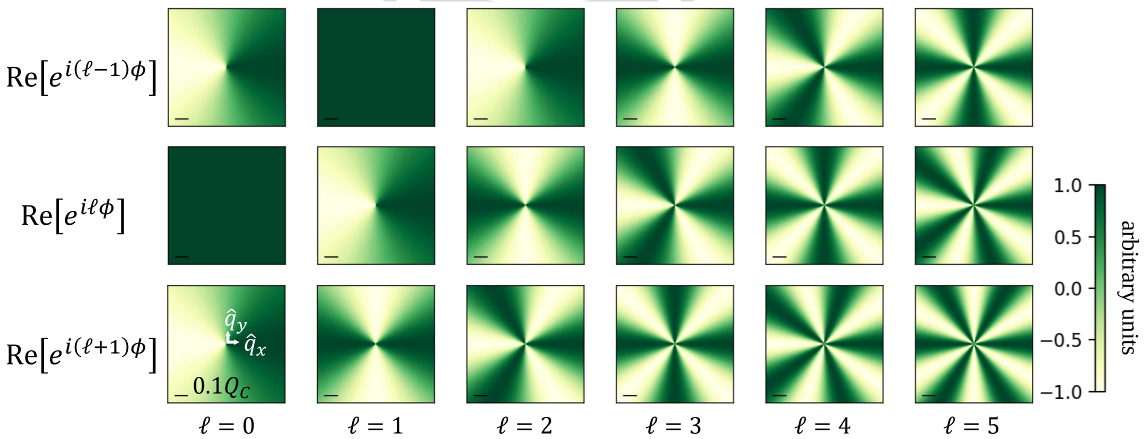

Figure 2.4：Phase distribution of different components of $\boldsymbol{\mathcal{A}}_{\ell,p}^{C}(\boldsymbol{q}_{\parallel})$ for different helicities. The scale bar indicates the considered 0.1 times light cone range for WSe$_2$-monolayers, $Q_c = 8.6\ \mu\mathrm{m}^{-1}$

2.2.3 SOI in the Magnitude of Vector Vortex Beam's (VVB) z Component

In Ref. [30], Oscar Javier Gomez Sanchez *et al.* pointed out that the information of total angular momentum (TAM) encoded in the phase can be lost when calculating optical transition rates. To retain and control this information, they introduced vector vortex beams (VVBs), which are constructed by superimposing two twisted light beams. The VVB state is expressed as a superposition through

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\|};\alpha,\beta)=\cos\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{q}_{\|})+e^{i\alpha}\sin\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}^{\sigma',\ell'}(\boldsymbol{q}_{\|}). \tag{2.30}$$

Here, the vector potential $\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{q}_{\|})=\boldsymbol{\mathcal{A}}^{C}_{\ell,p}(\boldsymbol{q}_{\|})$, the complex amplitude in Eq. (2.29), represents a LG beams characterized by SAM (σ) and OAM (ℓ). In this study, the radial mode index is set to $p=0$, as the primary focus remains on the interplay between SAM and OAM. The second term in the expression corresponds to another LG beams carrying a different combination of angular momentum (σ',ℓ').

The parameter α controls the relative phase between the two beams, while β determines their relative weight in the superposition. These two parameters can be interpreted as the azimuthal and polar angles on the Poincaré sphere (see Figure 2.5), providing a convenient geometric representation of the VVB state, which can be written as

$$|\sigma,\ell,\sigma',\ell';\alpha,\beta\rangle=\cos\left(\frac{\beta}{2}\right)|\sigma,\ell\rangle+e^{i\alpha}\sin\left(\frac{\beta}{2}\right)|\sigma',\ell'\rangle. \tag{2.31}$$

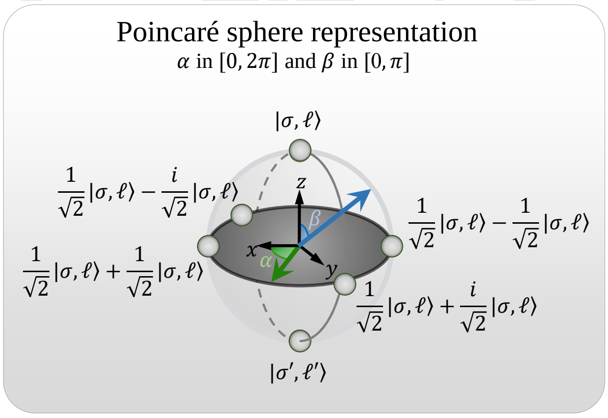

Figure 2.5: The VVB state represented on the Poincaré sphere, parameterized by the azimuthal angle α and polar angle β. Several representative combinations of α and β are illustrated on the sphere surface as examples.

Folowing Eq. (2.29), a VVB state also contains both transverse and longitudinal components, which can be written as

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\|};\alpha,\beta)=\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}_{\|}(\boldsymbol{q}_{\|};\alpha,\beta)\,\hat{\boldsymbol{\varepsilon}}+\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}_{z}(\boldsymbol{q}_{\|};\alpha,\beta)\,\hat{\boldsymbol{z}}. \tag{2.32}$$

The squared magnitudes of the transverse and longitudinal components are obtained as

$$\left|\boldsymbol{\mathcal{A}}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta)\right|^2 = \cos^2\left(\frac{\beta}{2}\right)\left|\tilde{F}_{\ell,p}(q_{\parallel})\right|^2 + \sin^2\left(\frac{\beta}{2}\right)\left|\tilde{F}_{\ell' p}(q_{\parallel})\right|^2 + \frac{1+\sigma\sigma'}{2}\sin\beta\,\tilde{F}_{\ell,p}(q_{\parallel})\tilde{F}^*_{\ell' p}(q_{\parallel})\cos\Big(\Delta\ell\,\phi_{\boldsymbol{q}_{\parallel}}+\alpha\Big)\,, \tag{2.33}$$

$$\left|\boldsymbol{\mathcal{A}}_{z}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta)\right|^2 = \frac{\sin^2\theta_{\boldsymbol{q}}}{2}\left[\cos^2\left(\frac{\beta}{2}\right)\left|\tilde{F}_{\ell,p}(q_{\parallel})\right|^2 + \sin^2\left(\frac{\beta}{2}\right)\left|\tilde{F}_{\ell' p}(q_{\parallel})\right|^2 + \sin\beta\,\tilde{F}_{\ell,p}(q_{\parallel})\tilde{F}^*_{\ell' p}(q_{\parallel})\cos\Big(\Delta J\,\phi_{\boldsymbol{q}_{\parallel}}+\alpha\Big)\right], \tag{2.34}$$

where the last term in each expression represents the interference between the two LG beams. For the transverse component, the interference vanishes when the two beams carry opposite SAM ($\sigma' = -\sigma$), since in that case $1+\sigma\sigma' = 0$. Here,

$$\Delta\ell = \ell' - \ell \tag{2.35}$$

is the difference in OAM between the two beams, while

$$\Delta J = (\ell' + \sigma') - (\ell + \sigma) \tag{2.36}$$

is the difference in TAM. In contrast to the transverse component, the longitudinal component explicitly contains the combined SAM–OAM contribution through ΔJ, reflecting its origin from the longitudinal field and encoding this structure in the amplitude rather than in the phase. In addition, the conditions for the disappearance of the azimuthal dependence on $\phi_{\boldsymbol{q}_{\parallel}}$ are different for the two components. For the transverse component, the azimuthal dependence vanishes when $\ell' = \ell$, whereas for the longitudinal component it disappears when $\Delta J = 0$.

For simplicity, we consider a VVB composed of two LG beams with opposite SAM and the same OAM absolute value ($\sigma' = -\sigma$ and $|\ell'| = |\ell|$). The components then reduce to

$$\left|\boldsymbol{\mathcal{A}}_{\parallel}^{\sigma,\ell,-\sigma,\ell}(\boldsymbol{q}_{\parallel};\alpha,\beta)\right|^2 = \left|\tilde{F}_{\ell,p}(q_{\parallel})\right|^2, \tag{2.37}$$

$$\left|\boldsymbol{\mathcal{A}}_{z}^{\sigma,\ell,-\sigma,\ell}(\boldsymbol{q}_{\parallel};\alpha,\beta)\right|^2 = \frac{\sin^2\theta_{\boldsymbol{q}}}{2}\left|\tilde{F}_{\ell,p}(q_{\parallel})\right|^2\left[1+\sin\beta\,\cos\Big(\Delta J\,\phi_{\boldsymbol{q}_{\parallel}}+\alpha\Big)\right]. \tag{2.38}$$

In this case, $\Delta J = -2\sigma$. The transverse component therefore exhibits an isotropic squared magnitude in angular spectrum and does not retain information about the SOI. By contrast, the longitudinal component reflects the TAM difference between the two LG beams through the interference term. Here, both beams share the same radial factor $\tilde{F}_{\ell,p}(q_{\parallel})$ for $|\ell'| = |\ell|$. As a result, the common radial factor can be factored out, allowing us to focus directly on the angular interference.

For a simple case $\sigma' = -\sigma = -1$ and $\ell' = \ell = 0$, Figure 2.6 shows that the squared magnitude of the longitudinal component exhibits TAM dependent azimuthal modulation as α and β are varied. In contrast, the transverse components retain the same intensity distribution over the Poincaré sphere, while the variation appears only in the polarization texture. Note that the factor $\sin^2\theta_{\boldsymbol{q}}$ in Eq. (2.38) vanishes near $\boldsymbol{q}_{\parallel} \approx 0$, producing a central null. As a result, the longitudinal component preserves a donut-shaped profile even when $\ell = \ell' = 0$ for both LG beams.

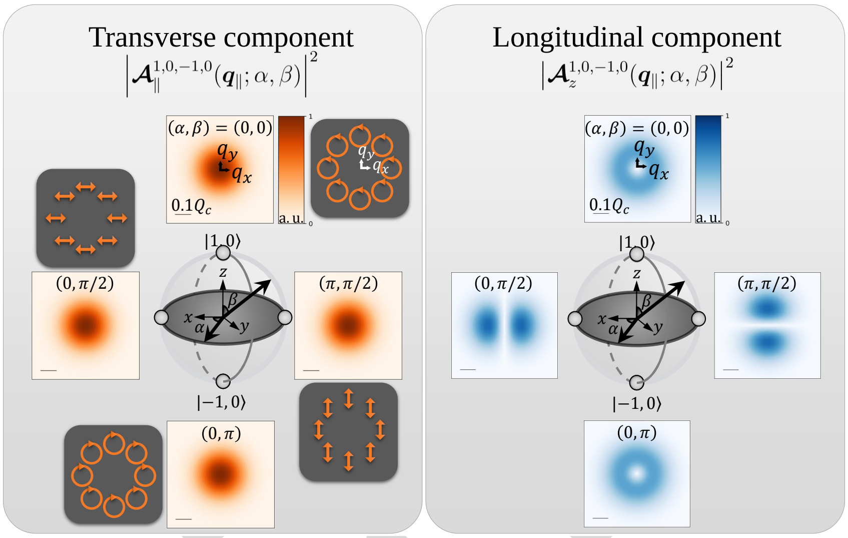

Figure 2.6 : Distribution (squared magnitude) of different components of $\boldsymbol{\mathcal{A}}_{\parallel}^{1,0,-1,0}(\boldsymbol{q}_{\parallel};\alpha,\beta)$ on the Poincaré sphere, together with the corresponding polarization states of the transverse components. The scale bar indicates the considered 0.1 times light cone range, $Q_c = 8.6\ \mu\text{m}^{-1}$

Motivated by this behavior, one may instead choose $\sigma' = -\sigma = -1$ and $\ell' = \ell = -1$, for which $\Delta J = (\ell' + \sigma') - (\ell + \sigma) = 0$. In this case, the longitudinal component becomes azimuthally uniform and can exhibit fully constructive or destructive interference over the entire angular spectrum when $(\alpha,\beta) = \left(0, \frac{\pi}{2}\right)$ or $\left(\pi, \frac{\pi}{2}\right)$, as shown in Figure 2.7.

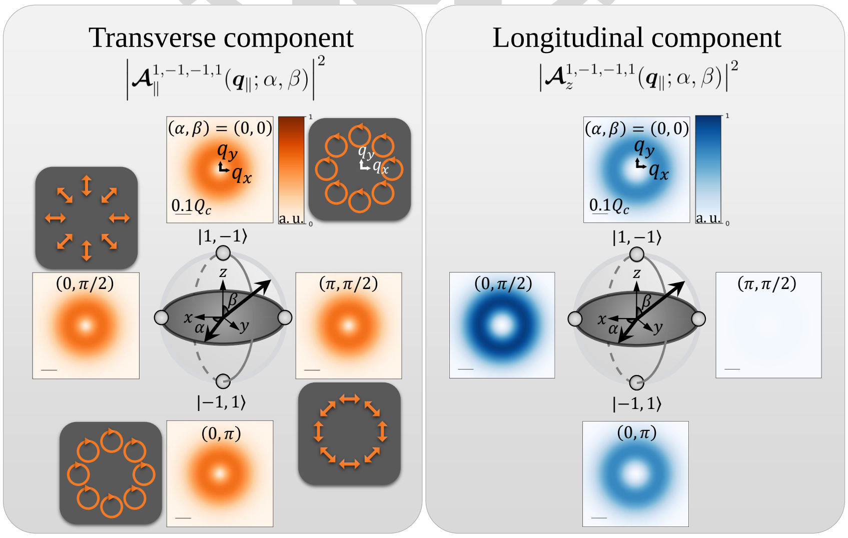

Figure 2.7 : Distribution (squared magnitude) of different components of $\boldsymbol{\mathcal{A}}_{\parallel}^{1,-1,-1,-1}(\boldsymbol{q}_{\parallel};\alpha,\beta)$ on the Poincaré sphere, together with the corresponding polarization states of the transverse components. The scale bar indicates the considered 0.1 times light cone range, $Q_c = 8.6\ \mu\text{m}^{-1}$

Chapter 3 Light-Matter Interaction

3.1 Theory of Light–Matter Interaction

3.1.1 Fermi's golden rule

In this section, we examine the interaction between light and matter using **Fermi's golden rule**. The transition rate for an optical excitation from the ground state $|GS\rangle$, with fully occupied valence bands, to an exciton state $|\Psi^X_{S,\boldsymbol{Q}}\rangle$ with band index S and center-of-mass momentum $\boldsymbol{Q}$ is given by

$$\Gamma = \frac{2\pi}{\hbar} \left| \langle \Psi^X_{S,\boldsymbol{Q}} | \hat{H}_I | GS \rangle \right|^2 \delta(E^X_{S,\boldsymbol{Q}} - \hbar\omega). \tag{3.1}$$

The delta function enforces energy conservation between the exciton energy $E^X_{S,\boldsymbol{Q}}$ and the photon energy $\hbar\omega$. Within the pseudospin model [6, 30, 41], the exciton state is written as

$$|\Psi^X_{S,\boldsymbol{Q}}\rangle = \frac{1}{\sqrt{\Omega}} \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \hat{c}^\dagger_{c,\boldsymbol{k}+\boldsymbol{Q}} \hat{h}^\dagger_{v,-\boldsymbol{k}} |GS\rangle. \tag{3.2}$$

Here, the particle operators $\hat{c}^\dagger_{c,\boldsymbol{k}}$ and $\hat{h}^\dagger_{v,-\boldsymbol{k}}$ create an electron in the conduction band with crystal momentum $\boldsymbol{k}$ and a hole in the valence band with crystal momentum $-\boldsymbol{k}$, respectively. The coefficient $\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ represents the amplitude of the electron–hole configuration for exciton band S, and Ω is the area of the TMD monolayer. The matrix element in Eq. (3.1) can be expressed, within the **rotating wave approximation** [42], as

$$\langle \Psi^X_{S,\boldsymbol{Q}} | \hat{H}_I | GS \rangle = \frac{1}{\sqrt{\Omega}} \sum_{vc\boldsymbol{k}} \Lambda^*_{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \langle \psi_{c,\boldsymbol{k}+\boldsymbol{Q}} | H_I | \psi_{v,\boldsymbol{k}} \rangle, \tag{3.3}$$

where, for weak fields, the light–matter interaction Hamiltonian is given by [6, 30]

$$\hat{H}_I = \frac{|e|}{2m_e} \boldsymbol{A}(\boldsymbol{r}, t) \cdot \boldsymbol{p},$$

with $\boldsymbol{A}(\boldsymbol{r}, t)$ the vector potential in the **Coulomb gauge**, $\boldsymbol{p}$ the momentum operator, and m_e (e) the electron mass (charge). If the vector potential is expanded in a **plane-wave basis** as in Eq. (2.25), the matrix element in Eq. (3.3) becomes

$$\begin{aligned} \langle \psi_{c,\boldsymbol{k}+\boldsymbol{Q}} | H_I | \psi_{v,\boldsymbol{k}} \rangle &= \frac{|e|}{2m_e} \sum_{\boldsymbol{q}_\parallel} \mathcal{A}(\boldsymbol{q}_\parallel) \left[\hat{\boldsymbol{\varepsilon}} \cdot \langle \psi_{c,\boldsymbol{k}+\boldsymbol{Q}} | e^{i\boldsymbol{q}_\parallel \cdot \boldsymbol{r}} \hat{\boldsymbol{p}} | \psi_{v,\boldsymbol{k}} \rangle \right] e^{i\omega t} \\ &\approx \frac{|e|}{2m_e} \sum_{\boldsymbol{q}_\parallel} \mathcal{A}(\boldsymbol{q}_\parallel) \left[\hat{\boldsymbol{\varepsilon}} \cdot \delta_{\boldsymbol{q}_\parallel, \boldsymbol{Q}} \langle \psi_{c,\boldsymbol{k}} | \hat{\boldsymbol{p}} | \psi_{v,\boldsymbol{k}} \rangle \right] e^{i\omega t}, \end{aligned} \tag{3.4}$$

where the approximation follows from the **electric dipole approximation**. In this approximation, the Bloch state wave functions are written as $\psi_{n,\boldsymbol{k}}(\boldsymbol{r}) = u_{n,\boldsymbol{k}}(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, and the plane-wave components combine to yield the Kronecker delta $\delta_{\boldsymbol{q}_\parallel,\boldsymbol{Q}}$, enforcing the equality between the in-plane wavevector and the exciton center-of-mass momentum.

Substituting this result back, the matrix element in Eq. (3.3) [6, 30] becomes

$$\langle \Psi^X_{S,\boldsymbol{Q}}|\hat{H}_I|GS\rangle = \frac{E_g}{2i\hbar}\mathcal{A}(\boldsymbol{Q})\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{X*}_{S,\boldsymbol{Q}}\right)e^{i\omega t}, \tag{3.5}$$

where $\boldsymbol{D}^X_{S,\boldsymbol{Q}}$ is the exciton transition dipole moment, superscript * denotes complex conjugation, and E_g is the band gap at the K/K' valleys. This expression shows that optical excitation is governed by the alignment between the light polarization and the transition dipole moment. The transition rate is therefore proportional to both the magnitude of the transition dipole moment and the magnitude of the vector potential. Under the excitaion of VVBs, the optical matrix element is defined as

$$\begin{aligned}\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{S,\boldsymbol{Q}} &\equiv \frac{1}{\sqrt{\Omega}}\sum_{vc\boldsymbol{k}}\Lambda^*_{S,\boldsymbol{Q}}(vc\boldsymbol{k})\left\langle \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}\left|\frac{|e|}{2m_e}\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\cdot\boldsymbol{p}\right|\psi_{v,\boldsymbol{k}}\right\rangle \\ &\approx \frac{E_g}{2i\hbar}\mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{X*}_{S,\boldsymbol{Q}}\right),\end{aligned} \tag{3.6}$$

where the approximation follows from the electric dipole approximation. The corresponding Fermi's golden rule can then be written as [30]

$$\Gamma^{\sigma,\ell,\sigma',\ell'}_{S,\boldsymbol{Q}} = \frac{2\pi}{\hbar}\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{S,\boldsymbol{Q}}\right|^2\rho(\hbar\omega = E^X_{S,\boldsymbol{Q}}) \propto \left|\mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^X_{S,\boldsymbol{Q}}\right)\right|^2. \tag{3.7}$$

The transition dipole moment, obtained from density functional theory combined with the Bethe–Salpeter equation, is given in the small-$\boldsymbol{Q}$ limit by Ref [6, 30]:

Valley excitons (bright)

$$\begin{aligned}\boldsymbol{D}^X_{K,\boldsymbol{Q}} &= \frac{D^X_B}{\sqrt{2}}(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}}), \\ \boldsymbol{D}^X_{K',\boldsymbol{Q}} &= \frac{D^X_B}{\sqrt{2}}(\hat{\boldsymbol{x}}-i\hat{\boldsymbol{y}})\end{aligned} \tag{3.8}$$

Valley-mixed excitons (bright)

$$\begin{aligned}\boldsymbol{D}^X_{B+,\boldsymbol{Q}} &= \frac{D^X_B}{\sqrt{2}}\left(\cos\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{x}}+\sin\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{y}}\right), \\ \boldsymbol{D}^X_{B-,\boldsymbol{Q}} &= \frac{iD^X_B}{\sqrt{2}}\left(-\sin\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{x}}+\cos\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{y}}\right)\end{aligned} \tag{3.9}$$

Gray excitons

$$\boldsymbol{D}^X_{G,\boldsymbol{Q}} = D^X_G\,\hat{\boldsymbol{z}}. \tag{3.10}$$

Here, the subscript symbols K, K', $B\pm$, and G denote the K valley, K' valley, upper/ lower band, and gray exciton, respectively. The magnitude of the dipole moment of BXs (GX) is D^X_B (D^X_G) [6, 30]. From Eq. (3.7), it becomes clear why a single LG beam, whose TAM is encoded purely in its phase, does not produce a transition rate that reflects SOI or TAM. This is because the transition rate depends on the squared magnitude of the vector potential, which removes all phase information.

3.2 Manifestation of SOI in Excitonic Optical Transitions

3.2.1 Optical Transition Pattern of Excitons

We now can utilize the formulation above to analyze the SOI in excitonic optical transition for both valley-mixed BXs and GX under VVBs. The transition rates can be evaluated with Eq. (2.33), Eq. (2.34), and Eq. (3.7), yielding:

$$\begin{aligned}
\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{B+,\boldsymbol{Q}}\right|^2 &= \left(\frac{E_g D^X_B}{2\sqrt{2}\hbar}\right)^2 \left|\cos(\beta/2)\tilde{F}_{\ell,p}(Q) + e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\sin(\beta/2)\tilde{F}_{\ell',p}(Q)\right|^2,\\
\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{B-,\boldsymbol{Q}}\right|^2 &= \left(\frac{E_g D^X_B}{2\sqrt{2}\hbar}\right)^2 \left|\sigma\cos(\beta/2)\tilde{F}_{\ell,p}(Q) + e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\sigma'\sin(\beta/2)\tilde{F}_{\ell',p}(Q)\right|^2,\\
\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{G,\boldsymbol{Q}}\right|^2 &= \left(\frac{E_g D^X_G}{2\sqrt{2}\hbar}\right)^2 \sin^2\theta_{\boldsymbol{Q}} \left|\cos(\beta/2)\tilde{F}_{\ell,p}(Q) + e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\sin(\beta/2)\tilde{F}_{\ell',p}(Q)\right|^2,
\end{aligned} \tag{3.11}$$

In this form, we observe that the SOI term $\Delta J = (\ell' + \sigma') - (\ell + \sigma)$ acts as an additional relative phase modulator with $\phi_{\boldsymbol{Q}}$ dependence. Comparing the cases of valley-mixed BXs and GXs, Eq. (3.5) shows that the spin index σ can enters the phase of the optical matrix element. Consequently, the $\phi_{\boldsymbol{Q}}$ dependence of the transition dipoles for valley-mixed BXs in Eq. (3.9) provides a mechanism for the emergence of ΔJ under excitation by a VVB state, whose polarization texture also depends on $\phi_{\boldsymbol{Q}}$. As a result, valley-mixed BXs can exhibit interference features analogous to those of GXs.

In contrast, for GXs, the SOI manifests directly through the interference term in Eq. (2.34). This originates from the fact that the transition dipole moment of GXs is oriented along the out-of-plane (z) direction and remains isotropic in the in-plane coordinates, making the interference the dominant mechanism responsible for revealing SOI with azimuthal structure. To explicitly reveal the interference effects, we expand the squared matrix elements for valley-mixed BXs as follows:

$$\begin{aligned}
\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{B+,\boldsymbol{Q}}\right|^2 = \left(\frac{E_g D^X_B}{2\sqrt{2}\hbar}\right)^2 \Big\{&\cos^2(\beta/2)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2(\beta/2)\left|\tilde{F}_{\ell',p}(Q)\right|^2\\
&+ \sin\beta\,\mathrm{Re}\left[\tilde{F}_{\ell,p}(Q)\tilde{F}^*_{\ell',p}(Q)e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\right]\Big\},\\
\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{B-,\boldsymbol{Q}}\right|^2 = \left(\frac{E_g D^X_B}{2\sqrt{2}\hbar}\right)^2 \Big\{&\cos^2(\beta/2)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2(\beta/2)\left|\tilde{F}_{\ell',p}(Q)\right|^2\\
&+\sigma\sigma'\sin\beta\,\mathrm{Re}\left[\tilde{F}_{\ell,p}(Q)\tilde{F}^*_{\ell',p}(Q)e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\right]\Big\}.
\end{aligned} \tag{3.12}$$

Similarly, for GXs, we obtain:

$$\begin{aligned}
\left|\tilde{M}^{\sigma,\ell,\sigma',\ell'}_{G,\boldsymbol{Q}}\right|^2 = \left(\frac{E_g D^X_G}{2\sqrt{2}\hbar}\right)^2 \sin^2\theta_{\boldsymbol{Q}}\Big\{&\cos^2(\beta/2)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2(\beta/2)\left|\tilde{F}_{\ell',p}(Q)\right|^2\\
&+ \sin\beta\,\mathrm{Re}\left[\tilde{F}_{\ell,p}(Q)\tilde{F}^*_{\ell',p}(Q)e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\right]\Big\}.
\end{aligned} \tag{3.13}$$

In general, the transition rates for BXs and GXs exhibit interference structures analogous to the familiar interference term of the VVB state in Eq. (2.34). In addition, the transition rates of BXs take the same functional form when $\sigma = \sigma'$. Below, we discuss valley-mixed BXs and GX separately, since the physical origins of the SOI term are different. This distinction also affects how the manipulation of the transition rate should be interpreted. Note that one cannot simply set $\sigma = 0$ to represent linearly polarized light in this formulation, because the exponential form has been taken to simplify the derivation of the final bracket in Eq. (3.7).

Valley-mixed Bright Excitons : Polarization-Matching

For valley-mixed BXs, Eq. (3.12) shows that using opposite circular polarizations ($\sigma' = -\sigma$) leads to opposite interference behavior for the two valley-mixed branches. In addition, the SOI term ΔJ, which does not explicitly appear in the transverse intensity profile of the VVB itself, emerges once the phase factor generated by the dot product between the polarization unit vector and the transition dipole moment is included in Eq. (3.5).

Motivated by the polarization texture of the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$, shown in Figure 2.7, the condition $\Delta J = 0$ can be interpreted as a polarization-matching mechanism, illustrated in Figure 3.1. In particular, the polarization of the states

$$|1,-1,-1,1;\alpha=0,\beta=\pi/2\rangle \quad \text{and} \quad |1,-1,-1,1;\alpha=\pi,\beta=\pi/2\rangle$$

matches the directions of the exciton transition dipoles D_B^X and D_B^X, respectively (see Figure 3.1). This enables selective excitation of the upper and lower exciton bands with the relative phase α as shown in Figure 3.2.

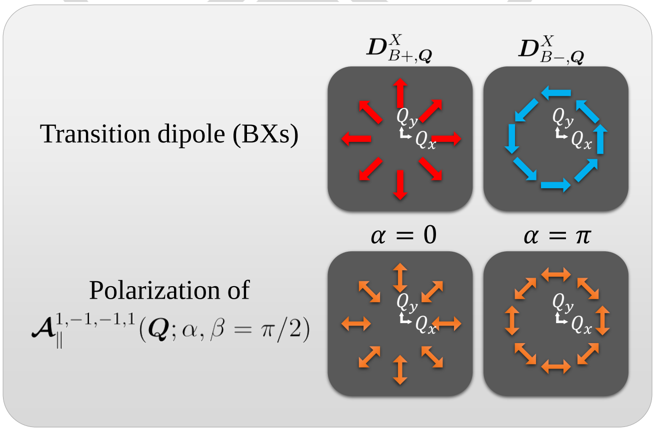

Figure 3.1 : Illustration of the transition dipole moments of valley-mixed BXs [6], and their matching with the transverse polarization of the VVB states $|1,-1,-1,1;\alpha=0,\beta=\pi/2\rangle$ and $|1,-1,-1,1;\alpha=\pi,\beta=\pi/2\rangle$ in momentum space.

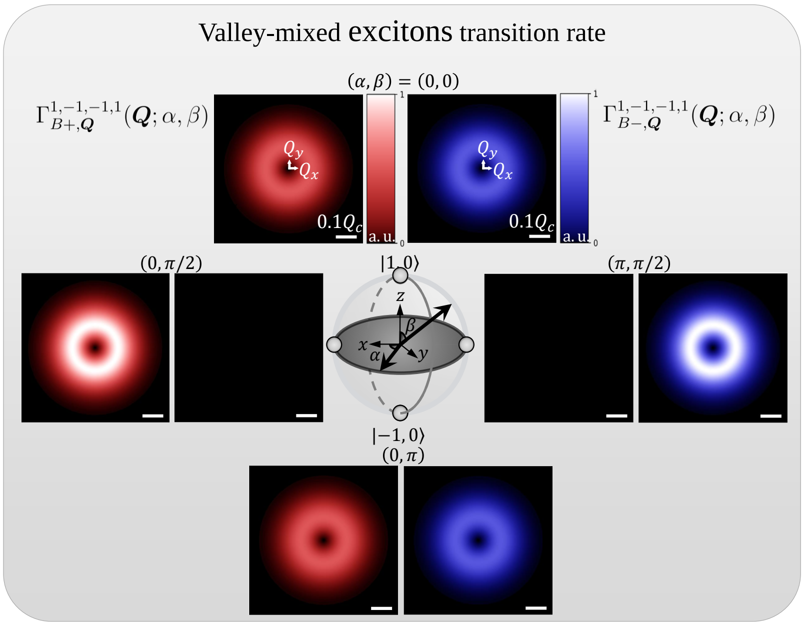

Figure 3.2：Normalized transition rate of valley-mixed BXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space. The scale bar indicates the considered 0.1 times light cone range for WSe$_2$-monolayers, $Q_c = 8.6\ \mu\text{m}^{-1}$

However, since the valley splitting between valley-mixed BXs is only a few meV and is typically regarded as spectrally unresolvable [43], direct experimental verification of selective excitation may be difficult. Therefore, we consider the transition rate for the total BXs under the condition $(\sigma' = -\sigma)$, which is given by

$$\begin{aligned}\Gamma^{\sigma,\ell,-\sigma,\ell'}_{B\pm,\boldsymbol{Q}}(\alpha,\beta) &\propto \sum_{S=B\pm}\left|\tilde{M}^{\sigma,\ell,-\sigma,\ell'}_{S,\boldsymbol{Q}}\right|^2 \\ &= \left(\frac{E_g D^X_B}{2\hbar}\right)^2\left[\cos^2\left(\frac{\beta}{2}\right)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2\left(\frac{\beta}{2}\right)\left|\tilde{F}_{\ell',p}(Q)\right|^2\right].\end{aligned} \tag{3.14}$$

The formulation shows the cancellation of the two interference terms, leading to the disappearance of both the α dependence and the ΔJ dependence. The resulting distribution is shown in Figure 3.3, which exhibits an identical donut-shaped profile over the Poincaré sphere.

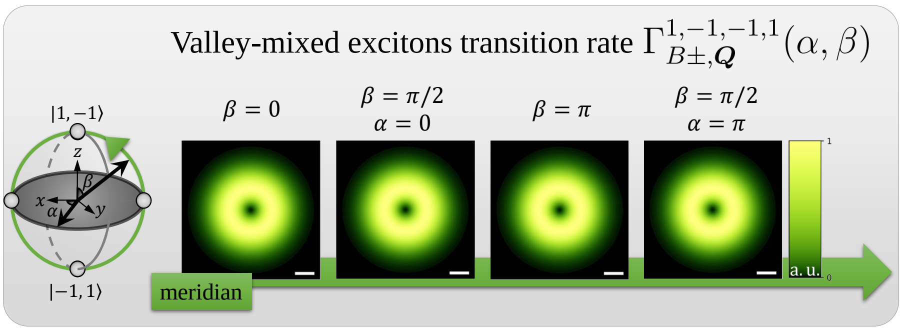

Figure 3.3：Normalized transition rate of total BXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space.

It should be noted that the condition $\Delta J = 0$ is not unique to this case. It can also be satisfied when $\ell' = \ell$ together with $\sigma' = \sigma$. However, in that situation the polarization-matching mechanism is absent. Instead, selective excitation is achieved simply by tuning the relative superposition parameters, effectively switching the excitation channel on and off, as shown in Figure 3.4 and Figure 3.5. Consequently, the excitation behavior of the two BX branches becomes identical, consistent with Eq. (3.12).

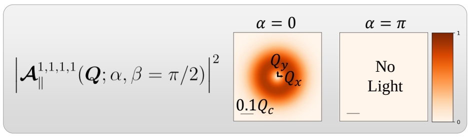

Figure 3.4：Distribution of the transverse component for the VVB state $|1,1,1,1;\alpha,\beta\rangle$ in momentum space.

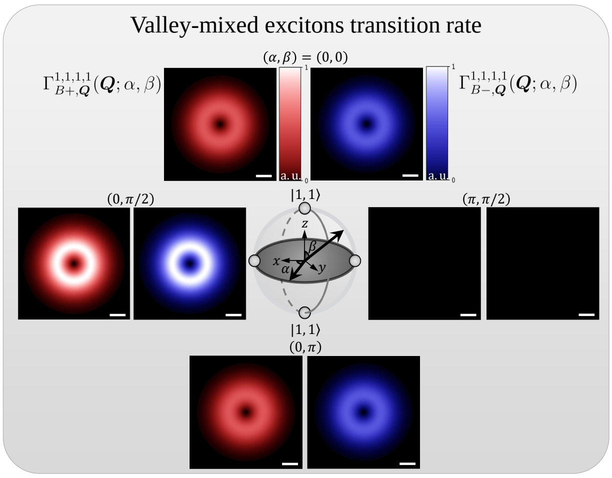

Figure 3.5：Normalized transition rate of valley-mixed BXs excited by the VVB state $|1,1,1,1;\alpha,\beta\rangle$ in momentum space.

Gray Excitons: Manifestation of SOI

For gray excitons, whose transition dipole moments possess only a z component, no additional phase factor is generated by the dot product between the polarization vector and the transition dipole moment in Eq. (3.5). Therefore, the SOI term ΔJ in Eq. (3.13) originates entirely from the longitudinal component of the VVB state in Eq. (2.34). As a result, GX provide an ideal probe of the SOI of light, since their transition rates directly reflect the intrinsic angular-momentum structure of the optical field. At the same time, they offer a possible route for exciting GX using paraxial structured beams.

Further exploit the intensity texture of the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$, shown in Figure 2.7. In particular, complete switching of the excitation on and off can be realized by tuning the relative phase between $\alpha = 0$ and $\alpha = \pi$ under excitation by the state

$$|1,-1,-1,1;\alpha,\beta = \pi/2\rangle,$$

as shown in Figure 3.6.

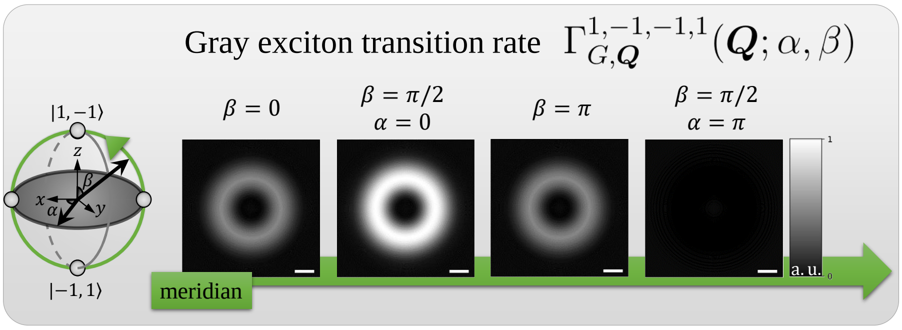

Figure 3.6：Normalized transition rate of GXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space.

Beyond complete switching, one may also use the case $\Delta J = 1$ to excite gray excitons with selected finite center-of-mass momentum. For example, consider the state

$$|1,-1,-1,2;\alpha,\beta = \pi/2\rangle,$$

for which

$$\Delta J = (\ell' + \sigma') - (\ell + \sigma) = 1.$$

First, the radial factor $\tilde{F}_{\ell p}(q_{||})$ of the two constituent LG beams are no longer identical, so the transition rates at $\beta = 0$ and $\beta = \pi$ (the north and south poles of the Poincaré sphere) is different. Figure 3.7 shows that the LG beam with larger ℓ produces a larger-radius donut-shaped transition-rate profile in the momentum space.

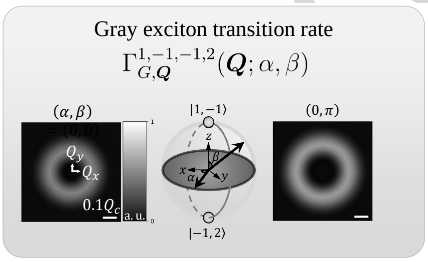

Figure 3.7：Normalized transition rate of GXs excited by the VVB state $|1,-1,-1,2;\alpha,\beta\rangle$ in momentum space at the north/south poles of the Poincaré sphere.

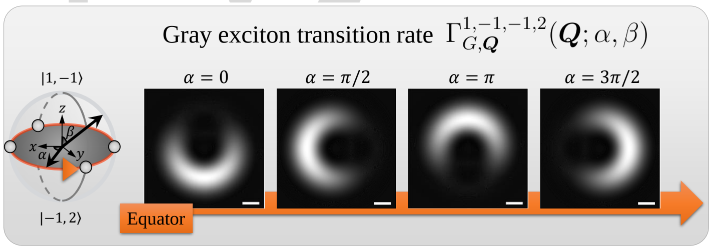

Figure 3.8：Normalized transition rate of GXs excited by the VVB state $|1,-1,-1,2;\alpha,\beta\rangle$ in momentum space along the equator of the Poincaré sphere. The scale and color bars are the same as in Figure 3.7.

Second, although the two LG beams do not fully overlap, the condition $\Delta J = 1$ still induces one full interference cycle in the azimuthal direction. Consequently, by tuning α, one can selectively excite GX within a finite range of center-of-mass momentum at $\beta = \pi/2$ (the equator of the Poincaré sphere), as shown in Figure 3.8.

3.2.2 Selective Excitation of Excitons

By summing the contributions in momentum space excited by the VVBs, we obtain the total transition rate, which represents the intensity of the material absorption. Under the paraxial approximation, where $\theta_{\boldsymbol{Q}} = \frac{Q}{\sqrt{Q^2+q_0^2}} \approx \frac{Q}{q_0}$ is taken to simplify the integral, the total transition rate can be evaluated as:

$$\Gamma_S^{\sigma,\ell,\sigma',\ell'} \equiv \sum_{\boldsymbol{Q}} \Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} = \frac{\Omega}{(2\pi)^2} \int_0^\infty \int_0^{2\pi} \Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \, Q \, \mathrm{d}\phi_{\boldsymbol{Q}} \, \mathrm{d}Q.$$

The total transition rates for the upper-band BXs, lower-band BXs, and GXs are obtained as

$$\begin{aligned}
\Gamma_{B+}^{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) &\propto \left|\tilde{M}_{B+}^{\sigma,\ell,\sigma',\ell'}\right|^2 \\
&= \frac{\Phi}{8}\left(\frac{E_g D_B^X}{a_0 \Omega \hbar}\right)^2 \left\{1 + \delta_{\Delta J,0} \sin\beta \cos\left[\frac{\pi}{2}(|\ell'| - |\ell|) + \alpha\right] \frac{\left(\frac{|\ell'|+|\ell|}{2}\right)!}{\sqrt{|\ell'|!|\ell|!}}\right\}, \\
\Gamma_{B-}^{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) &\propto \left|\tilde{M}_{B-}^{\sigma,\ell,\sigma',\ell'}\right|^2 \\
&= \frac{\Phi}{8}\left(\frac{E_g D_B^X}{a_0 \Omega \hbar}\right)^2 \left\{1 + \delta_{\Delta J,0}\sigma\sigma' \sin\beta \cos\left[\frac{\pi}{2}(|\ell'| - |\ell|) + \alpha\right] \frac{\left(\frac{|\ell'|+|\ell|}{2}\right)!}{\sqrt{|\ell'|!|\ell|!}}\right\}, \\
\Gamma_{G}^{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) &\propto \left|\tilde{M}_{G}^{\sigma,\ell,\sigma',\ell'}\right|^2 \\
&= \frac{\Phi}{4}\left(\frac{E_g D_G^X}{a_0 q_0 W_0 \Omega \hbar}\right)^2 \left\{\left(1 + \cos^2\frac{\beta}{2}|\ell| + \sin^2\frac{\beta}{2}|\ell'|\right)\right. \\
&\qquad \left. + \delta_{\Delta J,0} \sin\beta \cos\left[\frac{\pi}{2}(|\ell'| - |\ell|) + \alpha\right] \frac{\left(\frac{|\ell'|+|\ell|}{2}+1\right)!}{\sqrt{|\ell'|!|\ell|!}}\right\},
\end{aligned} \tag{3.15}$$

where the constant radiant power is $\Phi = (a_0 A_0^{\mathrm{LG}} W_0)^2$, and $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$.

At first glance, the SOI appears to be absent, since ℓ and σ are not explicitly coupled in the second term of all three expressions. However, the trigonometric factor can be reinterpreted as

$$\sin\beta\cos\left[\frac{\pi}{2}(|\ell'| - |\ell|) + \alpha\right] = \left(\mathrm{Re}\left[e^{i(|\ell'|-|\ell|)}\right], \mathrm{Im}\left[e^{i(|\ell'|-|\ell|)}\right], 0\right) \cdot \begin{pmatrix} \sin\beta\cos\alpha \\ \sin\beta\sin\alpha \\ \cos\beta \end{pmatrix}, \tag{3.16}$$

which reveals its origin as a phase correction between the complex components of two VVB states located on the poles of Poincaré sphere.

More specifically, the complex phase of the radial factor $\tilde{F}_{\ell,p}(Q)$ takes the form $(-i)^{|\ell|}$. As a result, the interference is governed by the difference in $|\ell|$, which introduces a phase shift across whole momentum space, together with the VVB state specified by (α, β).

At the same time, the contribution from SOI (or, for BXs, the polarization-matching mechanism) is encoded through $\delta_{\Delta J,0}$. When $\Delta J \neq 0$, the resulting azimuthal phase mismatch—such as that illustrated in Figure 3.8—leads to destructive averaging over momentum space, effectively given no net contribution to the total transition rate.

From this perspective, the factorial term in Eq. (3.15) can be interpreted as a geometric correction arising from the mismatch between the radial factors $\tilde{F}_{\ell,p}(Q)$ of the two LG beams. In particular, when $|\ell| = |\ell'|$, it reduces to simple constants, yielding 1 for BXs and $1 + |\ell|$ for GXs. These values are consistent with the corresponding prefactors in the non-interference contribution (i.e., the term independent of $\delta_{\Delta J,0}$).

As for the beam waist W_0, which appears in the denominator of the prefactor for the GX total transition rate, it reflects that smaller W_0 enhances the longitudinal field component and thus increases the excitation efficiency of GXs.

On the other hand, the total transition rate for the total BXs under the condition $(\sigma' = -\sigma)$ can be similarly written as

$$\Gamma_{B\pm}^{\sigma,\ell,-\sigma,\ell'}(\alpha,\beta) \propto \sum_{S=B\pm} \left| \tilde{M}_S^{\sigma,\ell,-\sigma,\ell'} \right|^2 = \frac{\Phi}{4} \left(\frac{E_g D_B^X}{a_0 \Omega \hbar} \right)^2 . \tag{3.17}$$

Therefore, one can exploit both Eq. (3.15) and Eq. (3.17) to analyze the optical transition rates. In particular, we focus on the equator of the Poincaré sphere ($\beta = \pi/2$), where the VVB corresponds to a maximal superposition of two LG beams. By varying the azimuthal angle α along the equator, we examine the dependence of the total transition rate on ℓ (with $|\ell| = |\ell'|$) using the VVB states $|1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, as shown in Figure 3.9.

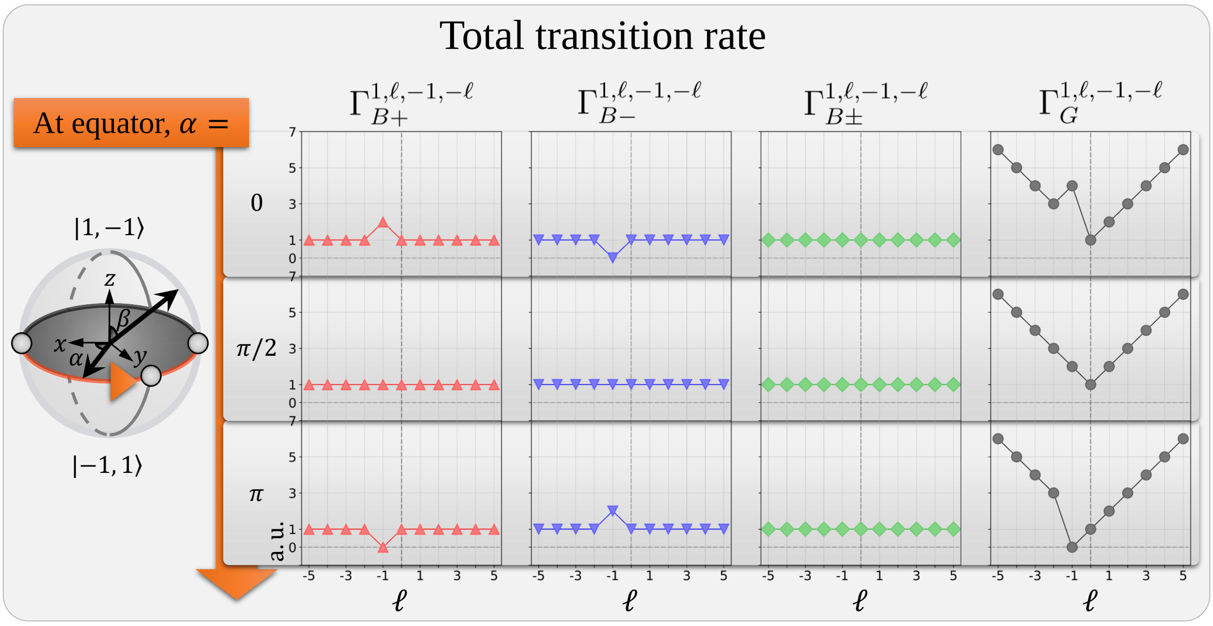

Figure 3.9：Normalized total transition rates of BXs, total BXs, and GX excited by a VVB state $|1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, for $\ell \in \mathbb{Z}$ with $\ell \in [-5, 5]$, and for azimuthal angles $\alpha = 0, \pi/2, \pi$.

In Figure 3.9, the total transition rate for each exciton exhibits a clear dependence on α

when the condition $\Delta J = 0$ is satisfied at $\ell = -1$. As shown within the same column, varying α from 0 to $\pi/2$ to π leads to modulation of the transition rate. This behavior is consistent with the results in Figure 3.2 and Figure 3.6, where the transition rate exhibits fully constructive or destructive interference over momentum space at $\alpha = 0$ or $\alpha = \pi$.

In contrast, the total transition rate of total BXs becomes independent of α and ℓ. This is due to the cancellation between the contributions of the upper and lower BX bands, as reflected in Figure 3.5.

Apart from this special cancellation, BXs generally exhibit independence from ℓ, unlike GXs. This difference originates from the additional $\boldsymbol{Q}$-dependent prefactor in the GX case, specifically $\sin\theta_{\boldsymbol{Q}} \approx Q/q_0$ in Eq. (3.13). Consequently, Eq. (3.15) shows a dependence on ℓ even in the absence of interference effects.

Motivated by the special properties of the VVB state at the equator of the Poincaré sphere, $|1, -1, -1, 1; \alpha, \beta = \pi/2\rangle$, one can achieve smooth and selective excitation over half of the equator ($\alpha = 0$ to π), as shown in Figure 3.10.

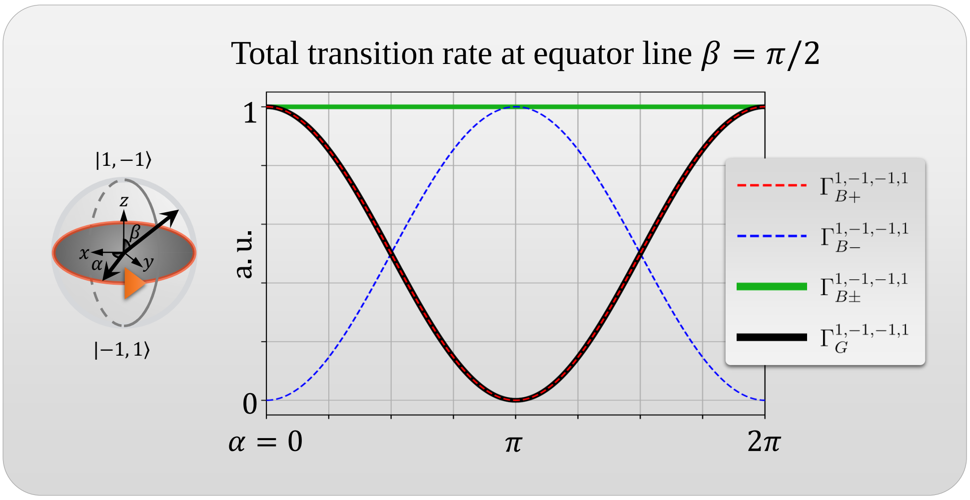

Figure 3.10：Normalized total transition rates of BXs, total BXs, and GXs excited by a VVB state $|1, -1, -1, 1; \alpha, \beta = \pi/2\rangle$ for azimuthal angles $\alpha = 0$ to 2π.

These observations originate from SOI associated with the longitudinal component of the LG beam, together with a polarization-matching mechanism. The former arises from the gauge transformation, specifically the second term in Eq. (2.23), as well as the intrinsic structure of the LG beam. The latter originates from the correspondence between the transition dipole structure of the upper and lower BX bands and the polarization textures of the VVB state. In both cases, Fermi's golden rule provides a theoretical framework for interpreting these results and establishes a basis for further studies involving different VVB configurations, excitonic species, and other comprehensive investigations.

Chapter 4 Wave Packets

4.1 Internal Structure of Excitons

4.1.1 Global Phase in Valley-Mixed Exciton

In this section, we establish the theoretical foundation for further studies of exciton wave packets. A wave packet of the particle refers to a superposition of momentum eigenstates, where the carrier wave (ripples) possesses a characteristic wavelength determined by the dominant momentum component, while the envelope describes the finite spatial extent of the particle. The velocity associated with the envelope, known as the group velocity, corresponds to the particle velocity in classical physics. From this perspective, the envelope of a wave packet provides a good intuition of the particle's position and motion [42, 44].

Using time-dependent perturbation theory [42], the time-dependent exciton state under weak photoexcitation can be expressed as a superposition of finite-momentum exciton (SFME) states [6],

$$|\Psi^X(t)\rangle = |GS\rangle + \sum_{S=B\pm}\sum_{\boldsymbol{Q}} \tilde{c}^{(1)}_{S,\boldsymbol{Q}}(t)e^{-i\omega_{S,\boldsymbol{Q}}t}|\Psi^X_{S,\boldsymbol{Q}}\rangle, \tag{4.1}$$

which represents a linear combination of the ground state $|GS\rangle$ and exciton states $|\Psi^X_{S,\boldsymbol{Q}}\rangle$ with center-of-mass (COM) momentum $\boldsymbol{Q}$. Other notations follow Eq. (3.2). Within first-order perturbation theory, the amplitude coefficients are given by

$$\tilde{c}^{(1)}_{S,\boldsymbol{Q}}(t) = \tilde{\gamma}_{S,\boldsymbol{Q}}(t)\left(\frac{\tilde{M}_{S,\boldsymbol{Q}}}{\hbar\omega_{S,\boldsymbol{Q}}}\right),$$

where time dependence coefficient $\tilde{\gamma}_{S,\boldsymbol{Q}}(t) \approx \tilde{\gamma}_{\boldsymbol{0}}(t)$, and the natural frequency is approximated as $\omega_{S,\boldsymbol{Q}} = E^X_{S,\boldsymbol{Q}}/\hbar \approx E^X_{S,\boldsymbol{0}}/\hbar = \omega_{\boldsymbol{0}}$. The amplitude coefficients can thus be separated into time-dependent and time-independent parts, where the latter contains the optical matrix element introduced in Eq. (3.6). For simplicity, we consider excitation by a single LG beam with radial mode index $p = 0$, such that the optical matrix element takes the form

$$\tilde{M}^{\hat{\varepsilon},\ell}_{S,\boldsymbol{Q}} \approx \frac{E_g}{2i\hbar}\mathcal{A}^{\hat{\varepsilon},\ell}(\boldsymbol{Q})\left(\hat{\boldsymbol{\varepsilon}}\cdot\boldsymbol{D}^{X*}_{S,\boldsymbol{Q}}\right). \tag{4.2}$$

Here, derivations analogous to those in the previous chapter—including the rotating-wave approximation and electric dipole approximation—are omitted for brevity. By adopting the polarization unit vector $\hat{\boldsymbol{\varepsilon}}$ instead of photon helicity σ, we relax the restriction to circular polarization. The vector potential is denoted by $\boldsymbol{\mathcal{A}}^{\hat{\varepsilon},\ell}(\boldsymbol{q}_\parallel) = \boldsymbol{\mathcal{A}}^C_{\ell,p}(\boldsymbol{q}_\parallel)$, which corresponds to the complex amplitude introduced in Eq. (2.29).

In the work of Guan-Hao Peng *et al.* [6], the analysis primarily focuses on the static part of the envelope function, which is defined by constructing the envelope function of the SFME state using only the amplitude coefficients,

$$F_S^{\hat{\varepsilon},\ell}(\boldsymbol{R}_c, t) = \frac{1}{\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{c}_{S,\boldsymbol{Q}}^{(1)}(t) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \tag{4.3}$$

To explicitly isolate the static contribution, we use the relation $F_S^{\hat{\varepsilon},\ell}(\boldsymbol{R}_c, t) \approx \tilde{\gamma}_{\boldsymbol{0}}(t)\, f_S^{\hat{\varepsilon},\ell}(\boldsymbol{R}_c)$, which defines the static envelope function as

$$f_S^{\hat{\varepsilon},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega_{\boldsymbol{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}_{S,\boldsymbol{Q}}^{\hat{\varepsilon},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \tag{4.4}$$

However, the valley-mixed exciton state obtained from the pseudo-spin model [6],

$$|\Psi_{\pm,\boldsymbol{Q}}^X\rangle = e^{i\delta(\boldsymbol{Q})} \left(e^{-i\phi_{\boldsymbol{Q}}}|\Psi_{K,\boldsymbol{Q}}^X\rangle \pm e^{i\phi_{\boldsymbol{Q}}}|\Psi_{K',\boldsymbol{Q}}^X\rangle\right)/\sqrt{2}, \tag{4.5}$$

naturally carries a $\boldsymbol{Q}$-dependent global phase $\delta(\boldsymbol{Q})$ arising from its internal structure. Consequently, the optical matrix element acquires the same phase factor,

$$\tilde{M}_{\pm,\boldsymbol{Q}}^{\hat{\varepsilon},\ell} = e^{i\delta(\boldsymbol{Q})} \left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}_{K,\boldsymbol{Q}}^{\hat{\varepsilon},\ell} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}_{K',\boldsymbol{Q}}^{\hat{\varepsilon},\ell}\right)/\sqrt{2}. \tag{4.6}$$

As a result, this $\boldsymbol{Q}$-dependent phase can modify the resulting static envelope function across momentum space, provided that the internal phase factors ($e^{\pm i\phi_{\boldsymbol{Q}}}$) partially compensate or reshape the global phase. This observation motivates a more careful examination of the internal structure of the exciton wave function and the explicit extraction of common factors.

4.1.2 Zeroth-Order Approximation

To address this issue without resorting to a numerically expensive construction of the exciton wave function, we instead extract its $\boldsymbol{Q}$–dependent phase structure analytically. We begin by considering the internal structure of the exciton wave function [45, 46],

$$\Psi_{S,\boldsymbol{Q}}^X(\boldsymbol{r}_e, \boldsymbol{r}_h) = \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{r}_e) \psi_{v,\boldsymbol{k}}^*(\boldsymbol{r}_h), \tag{4.7}$$

where $\psi_{c,\boldsymbol{k}}(\boldsymbol{r}_e)$ and $\psi_{v,\boldsymbol{k}}(\boldsymbol{r}_h)$ denote the single-particle Bloch wave functions for an electron in the conduction band and a hole in the valence band, respectively. Here, $\boldsymbol{r}_e$ and $\boldsymbol{r}_h$ are the corresponding electron and hole position vectors. The coefficient $\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ represents the amplitude of the electron–hole configuration for exciton band S.

For the purpose of extracting the COM momentum $\boldsymbol{Q}$-dependent phase term, it is convenient to rewrite the exciton wave function in terms of a weighted-average position vector $\boldsymbol{R}_c$[1]

[1] Although this weighted-average position vector is not strictly identical to the conventional COM coordinate in classical mechanics, we retain the subscript "c" to avoid confusion with the Bravais lattice vector $\boldsymbol{R}$.

and the relative coordinate $\boldsymbol{r}$:

$$\begin{gathered}\boldsymbol{R}_c = a\boldsymbol{r}_e + b\boldsymbol{r}_h, \qquad \boldsymbol{r} = \boldsymbol{r}_e - \boldsymbol{r}_h, \qquad a+b=1, \\ \boldsymbol{r}_e = \boldsymbol{R}_c + b\boldsymbol{r}, \qquad \boldsymbol{r}_h = \boldsymbol{R}_c - a\boldsymbol{r}.\end{gathered} \tag{4.8}$$

Using these relations, the exciton wave function can be rewritten as

$$\Psi^X_{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r})\psi^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - a\boldsymbol{r}). \tag{4.9}$$

In addition, Bloch's theorem states that, in a material with a periodic lattice structure—such as a two-dimensional TMD characterized by Bravais lattice vectors $\boldsymbol{R}$—the single-particle wave function satisfies [42, 46]

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}')e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{4.10}$$

where the wave function is expressed as the product of a cell-periodic function, $u_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}' + \boldsymbol{R})$, and a plane-wave component $e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}$ [2]. Using this property, the exciton wave function can be rewritten as

$$\begin{aligned}
&\Psi^X_{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) \\
&\Rightarrow \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q})\psi_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r})\psi^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) \\
&= \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q})\Big[u_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r})e^{i(\boldsymbol{k}+a\boldsymbol{Q})\cdot(\boldsymbol{R}_c+b\boldsymbol{r})}u^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r})e^{-i(\boldsymbol{k}-b\boldsymbol{Q})\cdot(\boldsymbol{R}_c-a\boldsymbol{r})}\Big] \\
&= \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\Big[u_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r})u^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r})\Big].
\end{aligned} \tag{4.11}$$

The first arrow is obtained by shifting crystal momentum by $-b\boldsymbol{Q}$, which allows the $\boldsymbol{Q}$-dependent phase originating from the plane-wave components to be written in the compact form $e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}$.

Finally, we adopt the convention

$$a = 1, \qquad b = 0,$$

which gives the exciton wave function in the form

$$\Psi^X_{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\left[u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - \boldsymbol{r})\right]. \tag{4.12}$$

Under this convention, the hole cell-periodic function becomes independent of $\boldsymbol{Q}$.

Further simplification can be achieved by performing a Taylor expansion around $\boldsymbol{Q} = \boldsymbol{0}$

[2] The vector $\boldsymbol{r}'$ denotes a general position vector and may represent the weighted-average, relative, electron, or hole position vector.

for the implicitly $\boldsymbol{Q}$-dependent quantities, namely $\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ and $u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)$. Explicitly,

$$\begin{aligned}
\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k}) &= \sum_{m=0}^{\infty}\frac{1}{m!}\left(\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\right)^m\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k}) &&= \Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k}) + \boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k}) + \cdots\,, \\
u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) &= \sum_{m=0}^{\infty}\frac{1}{m!}\left(\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\right)^m u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) &&= u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) + \boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) + \cdots\,.
\end{aligned} \tag{4.13}$$

Their product can therefore be expanded as

$$\begin{aligned}
\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) = &\underbrace{\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)}_{\text{zeroth order}} \\
&+ \underbrace{\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)\right] + \left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})\right]u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)}_{\text{first order}} \\
&+ \underbrace{\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})\right]\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)\right]}_{\text{second order}} \\
&+ \text{higher-order terms.}
\end{aligned} \tag{4.14}$$

Substituting this expansion into Eq. (4.12), the exciton wave function can be decomposed into contributions from different orders of $\boldsymbol{Q}$,

$$\begin{aligned}
\Psi^X_{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) &= e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\Big\{\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)\Big\}u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}) \\
&= e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\Big[\Psi^{X(0)}_{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) + \Psi^{X(1)}_{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) + \Psi^{X(2)}_{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) + \cdots\Big] \\
&\approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}),
\end{aligned} \tag{4.15}$$

where the final approximation refers to the zeroth-order approximation [3] in the small-$\boldsymbol{Q}$ limit. Consequently, the $\boldsymbol{Q}$-dependent phase factor can be extracted as [47]

$$\Psi^X_{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\Psi^X_{S,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.16}$$

In addition, the valley exciton wave functions with zero COM momentum satisfy

$$\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) = \Psi^{X*}_{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}), \tag{4.17}$$

as derived in appendix A.3. Therefore, each wave function can be expressed in polar form as

$$\begin{aligned}
\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \left|\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right|\exp\Big[i\,\mathrm{Arg}\big(\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\big)\Big], \\
\Psi^X_{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \left|\Psi^X_{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right|\exp\Big[i\,\mathrm{Arg}\big(\Psi^X_{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\big)\Big].
\end{aligned} \tag{4.18}$$

[3] A similar expansion scheme is commonly employed in evaluating the electron–hole exchange interaction (EHEI) within the Bethe–Salpeter equation, where the first-order contribution is typically retained. Although this may appear inconsistent with the zeroth-order approximation adopted here, the zeroth-order contribution to the EHEI vanishes to leading order, thereby necessitating the inclusion of higher-order terms.

From the complex-conjugation relation above, the two wave functions have identical amplitudes while their phases are opposite in sign, i.e.,

$$\left|\Psi^X_{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| = \left|\Psi^X_{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right|, \qquad \mathrm{Arg}\left[\Psi^X_{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right] = -\mathrm{Arg}\left[\Psi^X_{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]. \quad (4.19)$$

4.2 Static Envelope Function for Excitons

4.2.1 Exciton Wave Packet

Valley Excitons

For valley excitons, Eq. (4.16) suggests

$$\Psi^X_{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\Psi^X_{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \quad (4.20)$$

Accordingly, considering the internal structure of the exciton states, the envelope function can be constructed as a weighted superposition of exciton momentum eigenstates in real space, with amplitude coefficients derived from first–order time–dependent perturbation theory as

$$F^{\hat{\boldsymbol{\varepsilon}},\ell}_{K/K'}(\boldsymbol{R}_c,\boldsymbol{r},t) = \sum_{\boldsymbol{Q}} \tilde{c}^{(1)}_{K/K',\boldsymbol{Q}}(t)e^{-i\omega_{K/K',\boldsymbol{Q}}t}\Psi^X_{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \quad (4.21)$$

Similarly, through the relation $F^{\hat{\boldsymbol{\varepsilon}},\ell}_S(\boldsymbol{R}_c,\boldsymbol{r},t) \approx \tilde{\gamma}_{\mathbf{0}}(t)e^{-i\omega_{\mathbf{0}}t}f^{\hat{\boldsymbol{\varepsilon}},\ell}_S(\boldsymbol{R}_c,\boldsymbol{r})$, the static envelope function is defined as

$$f^{\hat{\boldsymbol{\varepsilon}},\ell}_{K/K'}(\boldsymbol{R}_c,\boldsymbol{r}) = (\hbar\omega_{\mathbf{0}})^{-1}\,\Psi^X_{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\sum_{\boldsymbol{Q}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K/K',\boldsymbol{Q}}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \quad (4.22)$$

At this stage, the presence of the unknown function $\Psi^X_{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ arising from the definition of the envelope function prevents further simplification. In the following, we address this issue by introducing physically motivated conditions under which its contribution can be regarded as negligible.

To this end, we approximate the neglect of both the constant prefactor $(\hbar\omega_{\mathbf{0}})^{-1}$ and the contribution from the zero COM exciton wave function $\Psi^X_{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})$. The former is omitted since we are primarily concerned with the spatial profile of the probability distribution, while the latter contains internal structure that can be analyzed separately.

The internal structure of the zero COM exciton wave function is given by Eq. (4.12) as

$$\Psi^X_{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) = \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\Lambda_{S,\mathbf{0}}(vc\boldsymbol{k})u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}). \quad (4.23)$$

We analyze this expression by separating it into three components: the phase factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, the coefficient $\Lambda_{S,\mathbf{0}}(vc\boldsymbol{k})$, and the cell-periodic functions $u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)$ and $u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r})$.

First, the phase factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ can be safely neglected by focusing on the case where the electron and hole spatially overlap, i.e., $\boldsymbol{r} = \boldsymbol{0}$. Beyond simplifying the analysis, this constraint physically corresponds to the configuration required for an exciton to undergo radiative recombination.

Second, the variation of the coefficient $\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})$ with respect to $\boldsymbol{k}$ can be estimated using the two-dimensional hydrogen model [48]. For the $1s$ exciton state, it is given by

$$\Lambda_{1s,\boldsymbol{0}}(vc\boldsymbol{k}) = \frac{\sqrt{2\pi}a_B^*}{\left[1+\left(\frac{a_B^*\boldsymbol{k}}{2}\right)^2\right]^{3/2}}, \tag{4.24}$$

where the effective Bohr radius is $a_B^* \approx 1\,\mathrm{nm}$. As shown in Figure 4.1, the coefficient $\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})$ remains nearly constant within the light cone region $\boldsymbol{Q}_c$, and is therefore expected to have a negligible influence on the summation.

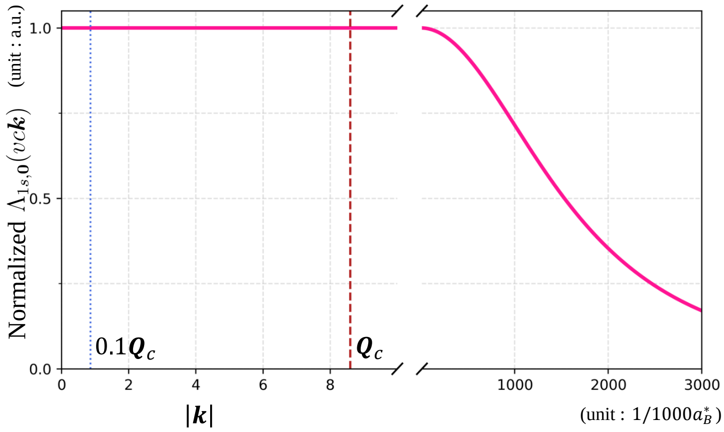

Figure 4.1 : Normalized coefficient $\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})$ along $\boldsymbol{k}$ (magenta), where the vertical dashed lines indecate the 0.1 and 1 times light cone range $\boldsymbol{Q}_c$ (blue and red). The light cone range $\boldsymbol{Q}_c = 8.6\mu\mathrm{m}^{-1}$ is considered for WSe_2. The maximun value is corresponding to $\sqrt{2\pi}a_B^*$

Finally, the cell-periodic functions satisfy $u_{n,\boldsymbol{k}}(\boldsymbol{R}_c) = u_{n,\boldsymbol{k}}(\boldsymbol{R}_c + \boldsymbol{R})$. Since the lattice constant (on the order of Å) is much smaller than the optical beam waist ($W_0 \sim 1.5\,\mu\mathrm{m}$), the variation of the cell-periodic functions is confined within a unit cell and does not significantly affect the overall spatial profile. From another perspective, neglecting the cell-periodic functions corresponds to describing the envelope function at discrete lattice sites, which does not qualitatively alter the result.

With the condition obtained above, and adopting Born's statistical interpretation [42], the probability density can be estimated from the absolute square of the static envelope function. In particular, for $\boldsymbol{r} = \boldsymbol{0}$,

$$\left|f^{\hat{\boldsymbol{\varepsilon}},\ell}_{K/K'}(\boldsymbol{R}_c, \boldsymbol{r} = \boldsymbol{0})\right|^2 \propto \left|\sum_{\boldsymbol{Q}} \tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K/K',\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\right|^2. \tag{4.25}$$

Valley-mixed Excitons

For valley–mixed excitons, the corresponding wave functions can be constructed within the pseudo–spin model [6] from the valley exciton wave functions. Notably, Eq. (4.16) remains applicable in this case. One obtains

$$
\begin{aligned}
\Psi^X_{\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) &\approx \frac{1}{\sqrt{2}}\left[e^{-i\phi_{\boldsymbol{Q}}}\Psi^X_{K,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \pm e^{i\phi_{\boldsymbol{Q}}}\Psi^X_{K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r})\right] \\
&\approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\Psi^X_{\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}).
\end{aligned}
\tag{4.26}
$$

Following Eq. (4.21) and combining it with Eq. (4.26), the envelope function can be written as

$$
F^{\hat{\boldsymbol{\varepsilon}},\ell}_{\pm}(\boldsymbol{R}_c,\boldsymbol{r},t) = \sum_{\boldsymbol{Q}} \tilde{c}^{(1)}_{\pm,\boldsymbol{Q}}(t)e^{-i\omega_{\pm,\boldsymbol{Q}}t}\Psi^X_{\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \tag{4.27}
$$

Similarly, using the relation $F^{\hat{\boldsymbol{\varepsilon}},\ell}_{S}(\boldsymbol{R}_c,\boldsymbol{r},t) \approx \tilde{\gamma}_{\boldsymbol{0}}(t)e^{-i\omega_{\boldsymbol{0}}t}f^{\hat{\boldsymbol{\varepsilon}},\ell}_{S}(\boldsymbol{R}_c,\boldsymbol{r})$, the static envelope function is defined as

$$
f^{\hat{\boldsymbol{\varepsilon}},\ell}_{\pm}(\boldsymbol{R}_c,\boldsymbol{r}) = (\hbar\omega_{\boldsymbol{0}})^{-1}\sum_{\boldsymbol{Q}} \tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{\pm,\boldsymbol{Q}}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\Psi^X_{\pm,0}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.28}
$$

Here, $\Psi^X_{\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ is retained inside the summation because the $\phi_{\boldsymbol{Q}}$ dependence remains embedded in its internal structure. Using both Eq. (4.5) and Eq. (4.6), the $\boldsymbol{Q}$–dependent terms can be made explicit by rewriting the static envelope function in terms of the valley indices K and K':

$$
\begin{aligned}
&f^{\hat{\boldsymbol{\varepsilon}},\ell}_{\pm}(\boldsymbol{R}_c,\boldsymbol{r}) \\
&= (2\hbar\omega_{\boldsymbol{0}})^{-1}\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K,\boldsymbol{Q}} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K',\boldsymbol{Q}}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\left[e^{-i\phi_{\boldsymbol{Q}}}\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \pm e^{i\phi_{\boldsymbol{Q}}}\Psi^X_{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right] \\
&= (2\hbar\omega_{\boldsymbol{0}})^{-1}\Bigg[\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K,\boldsymbol{Q}} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K',\boldsymbol{Q}}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{-i\phi_{\boldsymbol{Q}}} \\
&\qquad \pm \Psi^X_{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K,\boldsymbol{Q}} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K',\boldsymbol{Q}}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{i\phi_{\boldsymbol{Q}}}\Bigg].
\end{aligned}
\tag{4.29}
$$

Furthermore, using the complex–conjugation relation in Eq. (4.19), the amplitudes of the wave functions can be extracted, yielding

$$
\begin{aligned}
f^{\hat{\boldsymbol{\varepsilon}},\ell}_{\pm}(\boldsymbol{R}_c,\boldsymbol{r}) &= (2\hbar\omega_{\boldsymbol{0}})^{-1}\left|\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| \\
&\quad\Bigg[e^{i\mathrm{Arg}\left[\Psi^X_{K,0}(\boldsymbol{R}_c,\boldsymbol{r})\right]}\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K,\boldsymbol{Q}} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K',\boldsymbol{Q}}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{-i\phi_{\boldsymbol{Q}}} \\
&\quad \pm e^{-i\mathrm{Arg}\left[\Psi^X_{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]}\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K,\boldsymbol{Q}} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{\hat{\boldsymbol{\varepsilon}},\ell}_{K',\boldsymbol{Q}}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{i\phi_{\boldsymbol{Q}}}\Bigg].
\end{aligned}
\tag{4.30}
$$

In this form, the phase factors $\left(e^{\pm i\mathrm{Arg}\left[\Psi^X_{K,0}(\boldsymbol{R}_c,\boldsymbol{r})\right]}\right)$ can no longer be neglected. This is because the summations explicitly depend on the $\boldsymbol{Q}$-dependent terms $e^{\pm i\phi_{\boldsymbol{Q}}}$, which originate from the construction of the valley–mixed exciton states and give rise to nontrivial interference effects in momentum space. As a result, a complete construction of the exciton wave function appears, at first sight, unavoidable.

Nevertheless, the valley exciton envelope functions obtained in Eq. (4.25) still capture essential features and exhibit a clear correspondence to the four sum terms appearing in Eq. (4.30). Motivated by this observation, in the remainder of this work we focus on simulating the valley exciton envelope functions and analyzing their characteristic features.

Simulations

We begin with a brief summary of the following simulations. Under excitation by a single LG beam, the vector potential in the Coulomb gauge is given in Eq. (2.29). Accordingly, it can be written as

$$\mathcal{A}^{\hat{\varepsilon},\ell}(\boldsymbol{Q}) \approx \tilde{F}_{\ell,p=0}(\boldsymbol{Q})e^{i\ell\phi_{\boldsymbol{Q}}}\hat{\boldsymbol{\varepsilon}} - \frac{\sin\theta_{\boldsymbol{Q}}}{\sqrt{2}}\tilde{F}_{\ell,p=0}(\boldsymbol{Q})e^{i(\sigma+\ell)\phi_{\boldsymbol{Q}}}\hat{\boldsymbol{z}}, \tag{4.31}$$

where the approximation follows from the paraxial condition adopted in Eq. (2.28). For valley excitons, only the transverse component contributes to the optical matrix element, yielding

$$\tilde{M}^{\hat{\varepsilon},\ell}_{S,\boldsymbol{Q}} \approx \frac{E_g}{2i\hbar}\mathcal{A}^{\hat{\varepsilon},\ell}(\boldsymbol{Q})\left(\hat{\varepsilon}\cdot\boldsymbol{D}^{X*}_{S,\boldsymbol{Q}}\right). \tag{4.2}$$

Here, the phase is modified through the dot product with the transition dipole moment. Finally, the probability of finding the exciton is estimated using the approximate form of the valley exciton wave packet,

$$\left|f^{\hat{\varepsilon},\ell}_{K/K'}(\boldsymbol{R}_c, \boldsymbol{r}=\boldsymbol{0})\right|^2 \propto \left|\sum_{\boldsymbol{Q}} \tilde{M}^{\hat{\varepsilon},\ell}_{K/K',\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\right|^2, \tag{4.25}$$

which captures the spatial profile of the exciton probability distribution. The following simulations illustrate the squared magnitude distributions of the transverse component of the LG beam, $\left|\tilde{F}_{\ell,p=0}(\boldsymbol{Q})\right|^2$, the optical matrix element, $\left|\tilde{M}^{\hat{\varepsilon},\ell}_{S,\boldsymbol{Q}}\right|^2$, and the resulting static envelope function, $\left|\sum_{\boldsymbol{Q}} \tilde{M}^{\hat{\varepsilon},\ell}_{K/K',\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\right|^2$. In this way, we can track how the spatial profile evolves across these stages and relate each feature to its underlying physical origin.

For topological charge $\ell = 0$ and ± 1, the Gaussian–like and doughnut–shaped structures (see Figure 4.2 and Figure 4.4) are preserved in both the optical matrix element and the resulting probability distribution (see Figure 4.3 and Figure 4.5). This behavior directly reflects the spatial structure of the LG beam in angular spectrum space, which exhibits corresponding profiles in real space as well (see Figure 2.2).

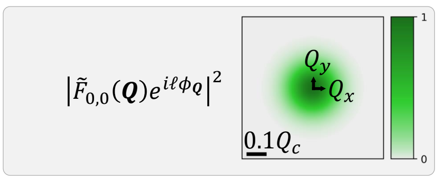

Figure 4.2：Normalized squared magnitude distribution of the transverse component of the LG beam ($\ell = 0$). The scale bar indicates 0.1 times the light cone range for WSe_2 monolayers, $\boldsymbol{Q}_c = 8.6\ \mu\text{m}^{-1}$.

In addition, the effect of strong spin–orbit coupling manifests as valley–selective optical transitions, which are also evident here: K and K' valley excitons are selectively excited by circularly polarized light with $\hat{\varepsilon} = \hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}$ and $\hat{\varepsilon} = \hat{\boldsymbol{x}} - i\hat{\boldsymbol{y}}$, respectively.

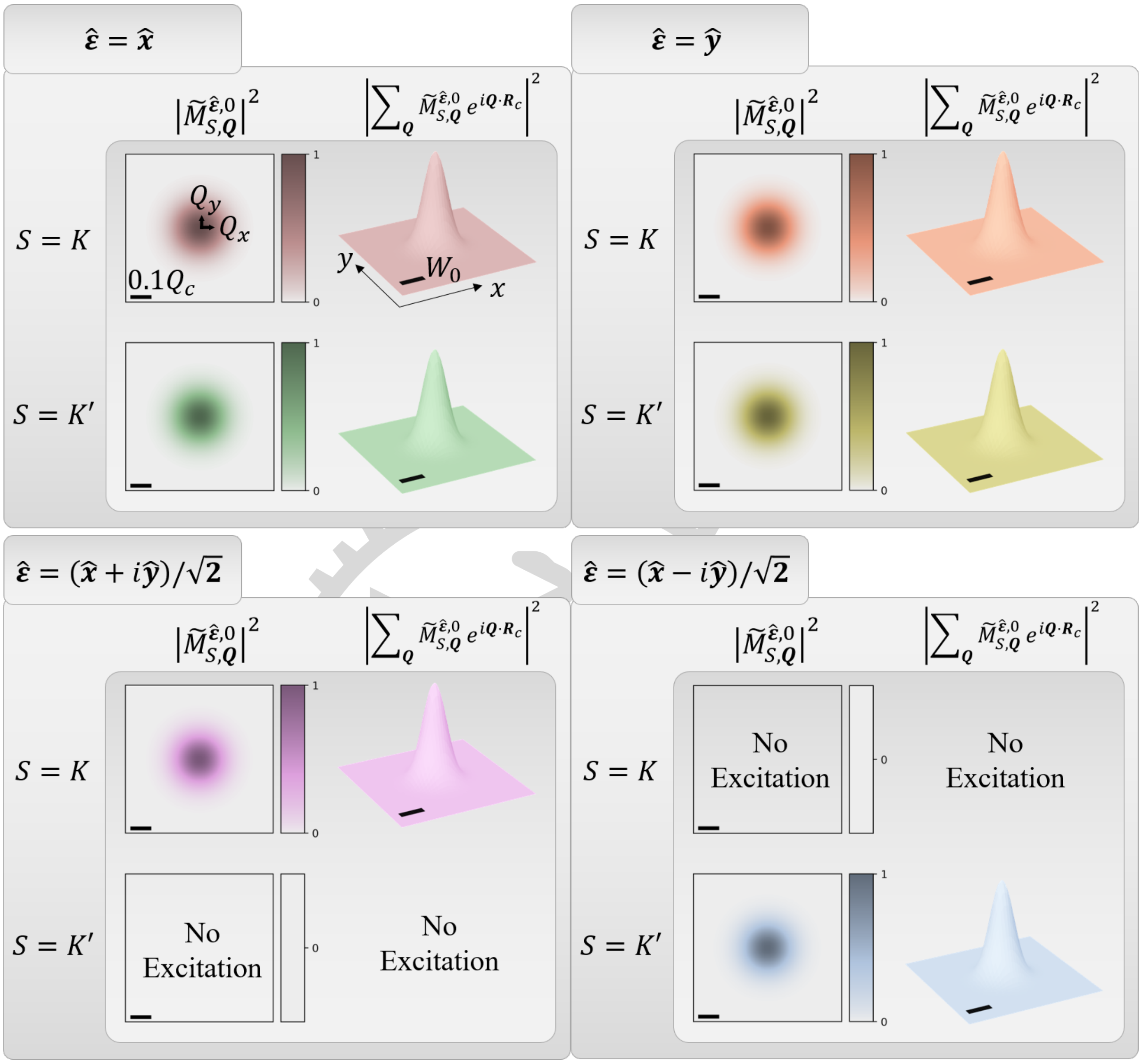

Figure 4.3 : Normalized squared magnitudes of the optical matrix element and the valley exciton wave packet excited by a LG beam ($\ell = 0$) with different polarization unit vectors $\hat{\varepsilon}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$, and the beam waist is $W_0 = 1.5\,\mu\text{m}^{-1}$.

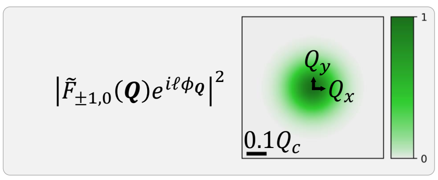

Figure 4.4 : Normalized squared magnitude distribution of the transverse component of the LG beam ($\ell = 1$). The scale bar indicates 0.1 times the light cone range for WSe_2 monolayers, $\boldsymbol{Q}_c = 8.6\,\mu\text{m}^{-1}$.

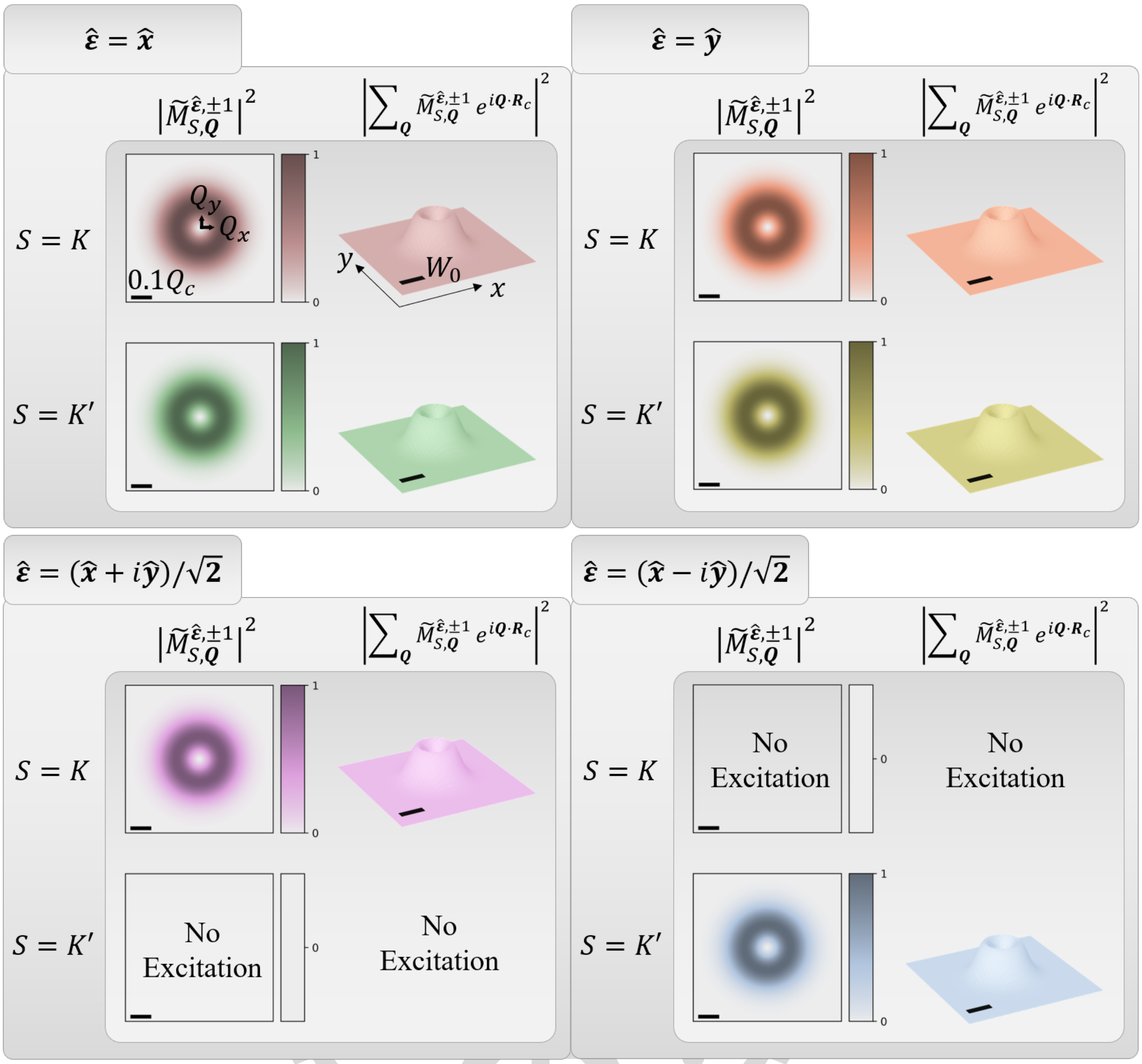

Figure 4.5：Normalized squared magnitudes of the optical matrix element and the valley exciton wave packet excited by a LG beam ($\ell = 1$) with different polarization unit vectors $\hat{\varepsilon}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$, and the beam waist is $W_0 = 1.5\,\mu\text{m}^{-1}$.

As a correspondence to the four sum terms appearing in Eq. (4.30), an artificial phase term could be added through

$$\left| f^{\hat{\varepsilon},\ell}_{K/K'}(\boldsymbol{R}_c, \boldsymbol{r} = \boldsymbol{0}) \right|^2 \propto \left| \sum_{\boldsymbol{Q}} \tilde{M}^{\hat{\varepsilon},\ell}_{K/K',\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} e^{-i\phi_{\boldsymbol{Q}}} \right|^2 \tag{4.32}$$

The estimation for whether there is a peak at the center can be obtained through

$$\begin{aligned} \left| f^{\hat{\varepsilon},\ell}_{K/K'}(\boldsymbol{R}_c = 0, \boldsymbol{r} = \boldsymbol{0}) \right|^2 &\propto \left| \sum_{\boldsymbol{Q}} \tilde{M}^{\hat{\varepsilon},\ell}_{K/K',\boldsymbol{Q}} e^{-i\phi_{\boldsymbol{Q}}} \right|^2 \\ &\propto \int_0^\infty \tilde{F}_{\ell,p=0}(Q) \left(\int_0^{2\pi} e^{i\ell\phi_{\boldsymbol{Q}}} e^{-i\phi_{\boldsymbol{Q}}}\, d\phi_{\boldsymbol{Q}} \right) dQ \\ &\propto \delta_{\ell,1}. \end{aligned} \tag{4.33}$$

For simplicity, we consider LG beams with polarization unit vectors $\hat{\varepsilon} = \hat{\boldsymbol{x}}$ and $\hat{\varepsilon} = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})/\sqrt{2}$. Similar behavior is observed for other polarization choices. In this case, the Gaussian–like and doughnut–shaped structures are no longer preserved in the resulting static envelope functions (see Figure 4.6). Instead, the profiles are effectively interchanged: a Gaussian–like beam excites a doughnut–shaped distribution, and vice versa. This qualitative change underscores the crucial role of the phase factors in Eq. (4.6), as the final summation is highly sensitive to these phase contributions.

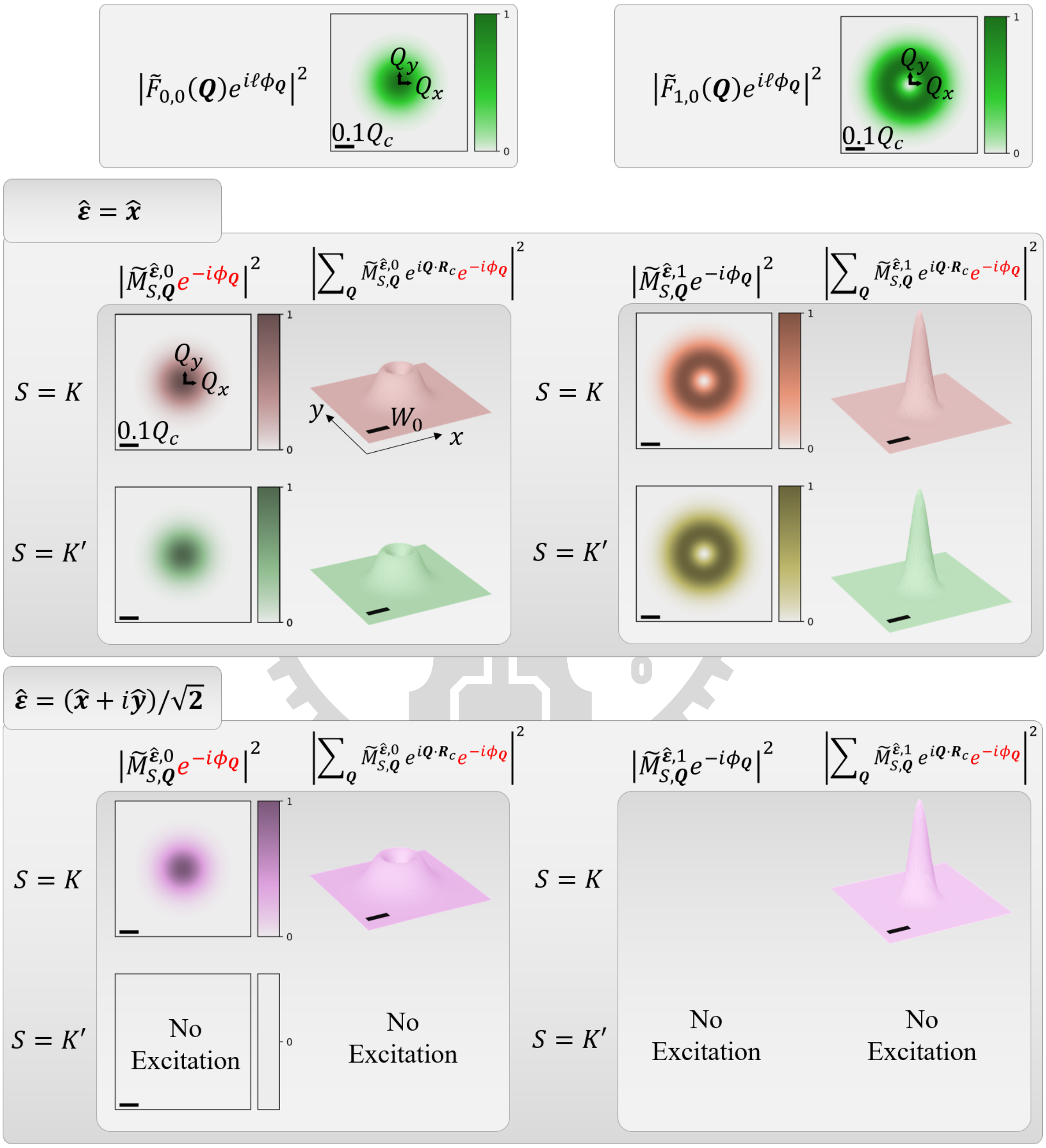

Figure 4.6 : Normalized squared magnitude distributions of the transverse component of the LG beam ($\ell = 0$), together with the normalized squared magnitudes of the optical matrix element and the resulting valley exciton wave packet for different polarization unit vectors $\hat{\varepsilon}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$, and the beam waist is $W_0 = 1.5\,\mu\mathrm{m}^{-1}$.

Chapter 5 Summary and outlook

In this study, we present a theoretical investigation of excitonic transitions in two–dimensional transition–metal dichalcogenides under photoexcitation by scalar and vector vortex beams.

In Chapter 2, we show that the well-defined orbital angular momentum (OAM) of $\ell\hbar$ carried by Laguerre–Gaussian (LG) beams is preserved under the paraxial approximation. Upon transformation to the Coulomb gauge, coupling with spin angular momentum (SAM) $\sigma\hbar$ naturally emerges. The adoption of the Coulomb gauge is justified within the framework of light–matter interaction, and the resulting spin–orbit interaction (SOI) term, $\sigma + \ell$, appears in the phase of the longitudinal component of LG beams in the angular spectrum representation. Furthermore, by superposing two LG beams to form vector vortex beams (VVBs), the SOI term is preserved as $\Delta J = (\sigma' + \ell') - (\sigma + \ell)$ in the amplitude of the longitudinal field. This quantity governs the variation of the relative phase along the azimuthal angle. Thus, when $\Delta J = 0$, the azimuthal structure vanishes, and the interference is fully controlled by the relative phase between the two LG beams, leading to either complete constructive or destructive outcomes.

In Chapter 3, Fermi's golden rule is employed to evaluate the transition rates of excitons, focusing on valley-mixed bright excitons (BXs) and gray excitons (GXs) under VVB excitation. Due to the approximately isotropic momentum-space distribution and out-of-plane orientation of the GX transition dipole, the transition rate directly reflects the longitudinal component of the VVB. As a result, selective excitation of GXs is achieved under the condition $\Delta J = 0$, corresponding to complete switching of the longitudinal field. On the other hand, valley-mixed BXs exhibits behavior similar to GXs. However, the condition $\Delta J = 0$ corresponds to matching between the BX transition dipole and the light polarization. Although selective excitation of valley-mixed BXs is theoretically possible, the few-meV valley splitting poses challenges for experimental verification.

In Chapter 4, first–order time–dependent perturbation theory is applied to construct a superposition of finite-momentum exciton (SFME) states, with the aim of determining the static envelope function of the exciton wave packet and examining the resulting spatial patterns. It is found that a global phase in valley-mixed BXs modifies the resulting envelope function, necessitating an explicit treatment of the center–of–mass momentum ($\boldsymbol{Q}$)–dependent phase structure of the exciton wave function. The extraction, based on zeroth-order and related approximations, is successful for valley excitons. In contrast, valley-mixed BXs introduce additional phase complexity originating from their internal structure. Simulations for valley excitons further demonstrate that phase factors play a crucial role in shaping the static envelope function.

Future work may extend this framework to other VVB configurations and incorporate full exciton wave functions. The fascinating interplay between structured light and matter enables further control of exciton dynamics, with potential applications in valleytronics and optoelectronic systems.

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Appendix

A.1 Wave Equations from Maxwell's Equations in Free Space

In classical electrodynamics, Maxwell's equations in vacuum are given by

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r},t) = \frac{\rho}{\epsilon_0} = 0, \tag{A.1.1}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r},t) = 0, \tag{A.1.2}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r},t)}{\partial t}, \tag{A.1.3}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r},t) = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \boldsymbol{E}(\boldsymbol{r},t)}{\partial t} = \frac{1}{c^2}\frac{\partial \boldsymbol{E}(\boldsymbol{r},t)}{\partial t}. \tag{A.1.4}$$

Here, $\boldsymbol{\nabla} \equiv \hat{\boldsymbol{x}}\frac{\partial}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial}{\partial z}$ denotes the spatial gradient operator in Cartesian coordinates. To reformulate the equations in terms of electromagnetic potentials, we first note that Eq. (A.1.2) implies that the magnetic field can be expressed as the curl of a vector potential $\boldsymbol{A}(\boldsymbol{r},t)$,

$$\boldsymbol{B}(\boldsymbol{r},t) = \boldsymbol{\nabla} \times \boldsymbol{A}(\boldsymbol{r},t). \tag{A.1.5}$$

Substituting Eq. (A.1.5) into Faraday's law (Eq. (A.1.3)) gives a curl-free expression, which allows the introduction of a scalar potential $\phi(\boldsymbol{r},t)$ such that

$$\boldsymbol{E}(\boldsymbol{r},t) = -\boldsymbol{\nabla}\phi(\boldsymbol{r},t) - \frac{\partial \boldsymbol{A}(\boldsymbol{r},t)}{\partial t}. \tag{A.1.6}$$

This representation automatically satisfies Eq. (A.1.2) and Eq. (A.1.3). Finally, Gauss's law (Eq. (A.1.1)) and Ampère–Maxwell law (Eq. (A.1.4)) can be rewritten as coupled wave equations for the scalar and vector potentials:

$$\nabla^2\phi(\boldsymbol{r},t) + \frac{\partial}{\partial t}\big(\boldsymbol{\nabla}\cdot\boldsymbol{A}(\boldsymbol{r},t)\big) = 0, \tag{A.1.7}$$

$$\nabla^2\boldsymbol{A}(\boldsymbol{r},t) - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}(\boldsymbol{r},t)}{\partial t^2} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}(\boldsymbol{r},t) + \frac{1}{c^2}\frac{\partial\phi(\boldsymbol{r},t)}{\partial t}\right) = 0. \tag{A.1.8}$$

A.2 Scalar Helmholtz Equation

We begin in the Lorenz gauge and assume a monochromatic-wave ansatz of the form $\boldsymbol{A}^L(\boldsymbol{r},t) = \mathrm{Re}\big[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\big]$. Substituting this expression into Eq. (2.8) yields the vector Helmholtz equation

$$\nabla^2\boldsymbol{A}^L(\boldsymbol{r}) + q_0^2\boldsymbol{A}^L(\boldsymbol{r}) = 0. \tag{A.2.1}$$

We then impose a transverse-field ansatz, $\boldsymbol{A}^L(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}\, A^L(\boldsymbol{r})$, where $\hat{\boldsymbol{\varepsilon}}$ is the polarization unit vector. Under this assumption, Eq. (A.2.1) reduces to a scalar equation for the amplitude $A^L(\boldsymbol{r})$,

$$\nabla^2 A^L(\boldsymbol{r}) + q_0^2 A^L(\boldsymbol{r}) = 0, \tag{A.2.2}$$

which is the scalar Helmholtz equation.

A.3 Exciton Wave Functions with Zero Center-of-Mass Momentum

In this appendix, we derive the relation between valley exciton wave functions with zero center-of-mass momentum at the high-symmetry valleys K and K'. Using the Bloch theorem, the corresponding exciton wave functions can be written as

$$\begin{aligned} \Psi^X_{K,\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) &\approx \psi_{c,\boldsymbol{K}}(\boldsymbol{R}_c)\psi^*_{v,\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r}) &&= u_{c,\boldsymbol{K}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r})e^{i\boldsymbol{K}\cdot\boldsymbol{r}}, \\ \Psi^X_{K',\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) &\approx \psi_{c,-\boldsymbol{K}}(\boldsymbol{R}_c)\psi^*_{v,-\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r}) &&= u_{c,-\boldsymbol{K}}(\boldsymbol{R}_c)u^*_{v,-\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r})e^{-i\boldsymbol{K}\cdot\boldsymbol{r}}, \end{aligned} \tag{A.3.1}$$

Here, the vectors $\boldsymbol{K}$ and $\boldsymbol{K}'$ denote the momentum-space position vectors of the K and K' valleys in reciprocal space, respectively. The single-particle Bloch wave functions satisfy[1]

$$\psi^*_{n,-\boldsymbol{K}}(\boldsymbol{r}') = \psi_{n,\boldsymbol{K}}(\boldsymbol{r}'). \tag{A.3.2}$$

Using the Bloch form of the single-particle wave function,

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}')e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{A.3.3}$$

Eq. (A.3.2) gives

$$u^*_{n,-\boldsymbol{K}}(\boldsymbol{r}') = u_{n,\boldsymbol{K}}(\boldsymbol{r}'). \tag{A.3.4}$$

Substituting this relation into Eq. (A.3.1), we obtain

$$\Psi^X_{K,\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) = \Psi^{X*}_{K',\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}), \tag{A.3.5}$$

which establishes the complex-conjugation relation between the valley exciton wave functions at $\boldsymbol{K}$ and $\boldsymbol{K}'$.

[1] The vector $\boldsymbol{r}'$ denotes a general position vector and may represent the weighted-average, relative, electron, or hole coordinate.